6 Execution, Testing and Results

```
STUD3#
STUD3#ping 10.1.10.1    ! Ping to Champs

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/32/32 ms
STUD3#
STUD3#
STUD3#
STUD3#ping 10.1.20.1    ! Ping to StAugustine

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.20.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/36/44 ms
STUD3#
STUD3#
STUD3#
STUD3#ping 10.1.1.1   ! Ping to FRX1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/33/36 ms
STUD3#
```

End-to-end IP connectivity verified via Frame Relay PVCs to FRX1, Champs Fleurs, and St. Augustine

```
FR1#show frame-relay route
Input Intf      Input Dlci      Output Intf     Output Dlci     Status
Serial0/1       200             Serial0/3       100             active
Serial0/1       201             Serial0/3       102             active
Serial0/2       300             Serial0/3       101             active
Serial0/3       100             Serial0/1       200             active
Serial0/3       101             Serial0/2       300             active
Serial0/3       102             Serial0/1       201             active
FR1#
```

Frame Relay switching table on FR1 showing all configured PVCs in active state, confirming that the four-switch Frame Relay network is correctly forwarding frames between STUD3, FRX1, Champs Fleurs, and St. Augustine

```
FR2#
FR2#show frame-relay route
Input Intf      Input Dlci      Output Intf     Output Dlci     Status
Serial0/0       500             Serial0/1       200             active
Serial0/1       200             Serial0/0       500             active
Serial0/1       201             Serial0/2       400             active
Serial0/2       400             Serial0/1       201             active
FR2#
```

FR2 switching table shows active routes for PVCs to FRX1 (DLCI 500) and to FR4 (DLCI 400)

```
FR3#
FR3#show frame-relay route
Input Intf      Input Dlci      Output Intf     Output Dlci     Status
Serial0/0       600             Serial0/2       300             active
Serial0/2       300             Serial0/0       600             active
FR3#
```

FR3 switching table shows active routes between FR1 and Champs Fleurs (DLCI 300/600)

```
FR4#
FR4#show frame-relay route
Input Intf      Input Dlci      Output Intf     Output Dlci     Status
Serial0/0       800             Serial0/2       400             active
Serial0/2       400             Serial0/0       800             active
FR4#
FR4#
```

FR4 switching table shows active routes between FR2 and St. Augustine (DLCI 400/800)

```
STUD1#ping 172.16.1.2    ! Ping GRE Tunnel to STUD3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/16/48 ms
STUD1#
```

GRE tunnel status on STUD1 showing line protocol is up and successful ping to STUD3's tunnel IP (172.16.1.2), confirming that the GRE VPN is operational

```
STUD3#ping 172.16.1.1    ! Ping GRE Tunnel to STUD1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/18/44 ms
STUD3#
```

GRE tunnel status on STUD3 showing line protocol is up and successful ping to STUD1's tunnel IP (172.16.1.1), confirming that the GRE VPN is operational

```
STUD3#
STUD3#show ip bgp summary   ! Display eBGP neighbor states and prefix counts
BGP router identifier 10.255.3.1, local AS number 65003
BGP table version is 23, main routing table version 23
21 network entries using 2457 bytes of memory
24 path entries using 1248 bytes of memory
8/7 BGP path/bestpath attribute entries using 992 bytes of memory
6 BGP AS-PATH entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 4841 total bytes of memory
BGP activity 22/1 prefixes, 28/4 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.1.1.1        4 65007      16      16       23    0    0 00:04:07       13
10.1.10.1       4 65005       8      15       23    0    0 00:04:03        3
10.1.20.1       4 65006       8      15       23    0    0 00:03:58        3
172.16.1.1      4 65001       0       0        0    0    0 never    Active
202.232.140.10  4 65008       0       0        0    0    0 never    Idle
STUD3#
STUD3#
```

eBGP Summary on STUD3 showing all Neighbors (FRX1, Champs, St. Aug, STUD1 via GRE) in Established state, Confirming Full BGP Mesh across the Core Network

```
STUD1#ping 202.232.140.10 ! Ping to FTPServer Target IP

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 202.232.140.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 656/710/740 ms
STUD1#
STUD1#
STUD1#traceroute 202.232.140.10 ! To FTPServer Target IP

Type escape sequence to abort.
Tracing the route to 202.232.140.10

  1 10.101.0.2 28 msec 32 msec 32 msec
  2 10.101.1.2 52 msec 60 msec 60 msec
  3 10.101.2.2 100 msec 68 msec 104 msec
  4 10.101.3.2 116 msec 132 msec 108 msec
  5 10.101.4.2 144 msec 176 msec 124 msec
  6 10.101.5.2 204 msec 192 msec 176 msec
  7 10.101.6.2 204 msec 216 msec 224 msec
  8 10.101.7.2 220 msec 280 msec 288 msec
  9 10.101.8.2 268 msec 260 msec 280 msec
 10 10.101.9.2 292 msec 336 msec 308 msec
 11 10.101.10.2 308 msec 344 msec 364 msec
 12 10.101.11.2 348 msec 368 msec 356 msec
 13 10.101.12.2 424 msec 368 msec 468 msec
 14 10.101.13.2 432 msec 428 msec 428 msec
 15 10.101.14.2 496 msec 500 msec 480 msec
 16 10.101.15.2 468 msec 480 msec 484 msec
 17 10.101.16.2 516 msec 572 msec 516 msec
 18 10.101.17.2 576 msec 564 msec 556 msec
 19 10.101.18.2 628 msec 588 msec 608 msec
 20 10.101.19.2 588 msec 612 msec 556 msec
 21 10.101.20.2 648 msec 608 msec 644 msec
 22 10.101.21.2 660 msec 708 msec 748 msec
 23 10.101.22.2 700 msec 784 msec 676 msec
STUD1#
```

Ping from STUD1 to FTPServer Target IP (202.232.140.10) / Traceroute from STUD1 to FTPServer (202.232.140.10) over the STUD1 hop chain

```
S4-HOP1#
S4-HOP1#ping 202.232.140.10 ! Ping FTPServer Target IP

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 202.232.140.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 556/632/704 ms
S4-HOP1#
S4-HOP1#
S4-HOP1#traceroute 202.232.140.10 ! To FTPServer Target IP

Type escape sequence to abort.
Tracing the route to 202.232.140.10

  1 10.104.1.2 96 msec 92 msec 72 msec
  2 10.104.2.2 124 msec 120 msec 120 msec
  3 10.104.3.2 148 msec 144 msec 120 msec
  4 10.104.4.2 148 msec 152 msec 148 msec
  5 10.104.5.2 176 msec 224 msec 172 msec
  6 10.104.6.2 212 msec 208 msec 176 msec
  7 10.104.7.2 208 msec 284 msec 224 msec
  8 10.104.8.2 212 msec 220 msec 276 msec
  9 10.104.9.2 256 msec 304 msec 300 msec
 10 10.104.10.2 272 msec 236 msec 308 msec
 11 10.104.11.2 364 msec 268 msec 328 msec
 12 10.104.12.2 332 msec 372 msec 332 msec
 13 10.104.13.2 360 msec 352 msec 328 msec
 14 10.104.14.2 412 msec 356 msec 360 msec
 15 10.104.15.2 392 msec 384 msec 528 msec
 16 10.104.16.2 464 msec 508 msec 412 msec
 17 10.104.17.2 448 msec 440 msec 444 msec
 18 10.104.18.2 492 msec 480 msec 448 msec
 19 10.104.19.2 540 msec 472 msec 504 msec
 20 10.104.20.2 548 msec 496 msec 496 msec
 21 10.104.21.2 500 msec 580 msec 564 msec
 22 10.104.22.2 584 msec 584 msec 560 msec
 23 10.104.23.2 592 msec 596 msec 528 msec
 24 10.104.24.2 544 msec 700 msec 564 msec
S4-HOP1#
```

Ping from S4-HOP1 to FTPServer Target IP (202.232.140.10) / Traceroute from S4-HOP1 to FTPServer (202.232.140.10) over the STUD4 hop chain

```
STUD2#
STUD2#
STUD2#ping 10.102.25.1 ! Ping to S2-HOP25

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.102.25.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 740/775/796 ms
STUD2#
STUD2#
STUD2#ping 202.232.140.10 ! Ping to FTPServer (Target IP)

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 202.232.140.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 776/782/796 ms
STUD2#
STUD2#
STUD2#traceroute 202.232.140.10  ! To FTPServer Target IP

Type escape sequence to abort.
Tracing the route to 202.232.140.10

  1 10.102.0.2 20 msec 32 msec 28 msec
  2 10.102.1.2 48 msec 60 msec 60 msec
  3 10.102.2.2 88 msec 92 msec 172 msec
  4 10.102.3.2 112 msec 132 msec 128 msec
  5 10.102.4.2 156 msec 152 msec 152 msec
  6 10.102.5.2 180 msec 164 msec 204 msec
  7 10.102.6.2 216 msec 216 msec 244 msec
  8 10.102.7.2 316 msec 240 msec 264 msec
  9 10.102.8.2 284 msec 272 msec 308 msec
 10 10.102.9.2 292 msec 324 msec 288 msec
 11 10.102.10.2 352 msec 340 msec 304 msec
 12 10.102.11.2 380 msec 392 msec 360 msec
 13 10.102.12.2 412 msec 380 msec 420 msec
 14 10.102.13.2 452 msec 436 msec 416 msec
 15 10.102.14.2 480 msec 420 msec 472 msec
 16 10.102.15.2 516 msec 504 msec 476 msec
 17 10.102.16.2 524 msec 496 msec 508 msec
 18 10.102.17.2 552 msec 592 msec 548 msec
 19 10.102.18.2 556 msec 600 msec 588 msec
 20 10.102.19.2 616 msec 632 msec 608 msec
 21 10.102.20.2 660 msec 620 msec 668 msec
 22 10.102.21.2 700 msec 712 msec 684 msec
 23 10.102.22.2 728 msec 716 msec 708 msec
 24 10.102.23.2 704 msec 740 msec 732 msec
 25 10.102.24.2 784 msec 764 msec 764 msec
 26 10.102.25.2 784 msec 836 msec 744 msec
STUD2#
```

Ping from STUD2 to S2-HOP25 & FTPServer Target IP (202.232.140.10) / Traceroute from STUD2 Router to FTPServer (202.232.140.10) over the STUD2 hop chain

```
SA-PC51> ping 202.232.140.10
84 bytes from 202.232.140.10 icmp_seq=1 ttl=230 time=775.901 ms
84 bytes from 202.232.140.10 icmp_seq=2 ttl=230 time=783.612 ms
84 bytes from 202.232.140.10 icmp_seq=3 ttl=230 time=801.235 ms
84 bytes from 202.232.140.10 icmp_seq=4 ttl=230 time=774.421 ms
84 bytes from 202.232.140.10 icmp_seq=5 ttl=230 time=817.119 ms

SA-PC51>
```

Ping from SA-PC51 (STUD4 PC1) to FTPServer Target IP (202.232.140.10)

```
STUD4#show ip nat translations
Pro Inside global      Inside local       Outside local      Outside global
icmp 10.104.0.1:11510  192.168.4.12:11510 202.232.140.10:11510 202.232.140.10:11510
icmp 10.104.0.1:12022  192.168.4.12:12022 202.232.140.10:12022 202.232.140.10:12022
icmp 10.104.0.1:12278  192.168.4.12:12278 202.232.140.10:12278 202.232.140.10:12278
icmp 10.104.0.1:12790  192.168.4.12:12790 202.232.140.10:12790 202.232.140.10:12790
icmp 10.104.0.1:13302  192.168.4.12:13302 202.232.140.10:13302 202.232.140.10:13302
STUD4#
STUD4#
STUD4#show ip nat statistics
Total active translations: 5 (0 static, 5 dynamic; 5 extended)
Outside interfaces:
  FastEthernet0/0, Serial0/0
Inside interfaces:
  FastEthernet1/0
Hits: 5  Misses: 5
CEF Translated packets: 10, CEF Punted packets: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list 1 interface FastEthernet0/0 refcount 5
Appl doors: 0
Normal doors: 0
Queued Packets: 0
STUD4#
```

NAT translations and statistics on STUD4

```
STUD1#
STUD1#show ntp status
Clock is unsynchronized, stratum 16, no reference clock
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**18
reference time is 00000000.00000000 (00:00:00.000 UTC Mon Jan 1 1900)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 0.00 msec, peer dispersion is 0.00 msec
STUD1#
STUD1#
STUD1#show ntp associations

      address         ref clock      st  when  poll reach  delay  offset    disp
 ~202.232.140.10   0.0.0.0          16     -    64     0     0.0    0.00  16000.
 * master (synced), # master (unsynced), + selected, - candidate, ~ configured
STUD1#
STUD1#
STUD1#
```

NTP status on STUD1 after configuring ntp server 202.232.140.10

```
FTPServer#
FTPServer#show ntp status   ! Verify FTPServer as NTP master
Clock is synchronized, stratum 1, reference is .LOCL.
nominal freq is 250.0000 Hz, actual freq is 250.0000 Hz, precision is 2**18
reference time is C0296093.383653D8 (02:06:11.219 UTC Fri Mar 1 2002)
clock offset is 0.0000 msec, root delay is 0.00 msec
root dispersion is 0.02 msec, peer dispersion is 0.02 msec
FTPServer#
```

Verification of FTPServer being the NTP Master

```
SA-PC52> show ip

NAME        : SA-PC52[1]
IP/MASK     : 192.168.4.12/24
GATEWAY     : 192.168.4.1
DNS         : 202.232.140.10
MAC         : 00:50:79:66:68:05
LPORT       : 10664
RHOST:PORT  : 127.0.0.1:10665
MTU:        : 1500

SA-PC52>
SA-PC52> ip dhcp
DDD
Can't find dhcp server

SA-PC52> dhcp
DDD
Can't find dhcp server
```

SA-PC52 (From STUD4 LAN) Failed attempt at Connecting to DHCP Server

```
STUD1#
STUD1#show access-list 100   ! Display the ACL applied to Tunnel0
Extended IP access list 100
    10 permit tcp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255 eq www
    20 permit tcp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255 eq 443
    30 permit tcp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255 eq bgp
    40 permit icmp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255
    50 permit udp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255 eq bootps
    60 permit udp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255 eq bootpc
    70 permit udp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255 eq ntp
    80 permit tcp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255 eq domain
    90 permit udp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255 eq domain
    100 permit tcp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255 eq 22
    110 deny ip any any (1806 matches)
STUD1#
STUD1#
STUD1#show ip interface Tunnel0   ! Verify ACL is applied inbound
Tunnel0 is up, line protocol is up
  Internet address is 172.16.1.1/30
  Broadcast address is 255.255.255.255
  Address determined by non-volatile memory
  MTU is 1476 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound  access list is 100
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachables are always sent
  ICMP mask replies are never sent
  IP fast switching is enabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP CEF switching is enabled
  IP CEF Feature Fast switching turbo vector
  IP multicast fast switching is enabled
  IP multicast distributed fast switching is disabled
  IP route-cache flags are Fast, CEF
  Router Discovery is disabled
  IP output packet accounting is disabled
  IP access violation accounting is disabled
  TCP/IP header compression is disabled
  RTP/IP header compression is disabled
  Policy routing is disabled
  Network address translation is disabled
  BGP Policy Mapping is disabled
  WCCP Redirect outbound is disabled
  WCCP Redirect inbound is disabled
  WCCP Redirect exclude is disabled
STUD1#
```

Access-list 100 on STUD1 showing the rules that permit traffic from STUD2's LAN (192.168.2.0/24) to STUD4's LAN (192.168.4.0/24) over the GRE tunnel, followed by a deny any

7 Summary, Issues and Limitations

Summary of Work Performed

This project successfully implemented a WAN infrastructure in GNS3, incorporating Frame Relay, eBGP, GRE VPN, NAT, DHCP, NTP, DNS, and ACLs. The topology included 4 student routers (STUD1-STUD4), 3 Frame Relay edge routers (Champs Fleurs, St. Augustine, FRX1), 4 Frame Relay switches (FR1-FR4), and an FTPServer providing DHCP, NTP, and DNS services. eBGP was configured across all routers using AS numbers 65001-65008. A GRE VPN tunnel was established between STUD1 and STUD3. Frame Relay was implemented with a square topology (FR1-FR2-FR4-FR3), with CIR limitations of 128 kbps/64 kbps between Champs Fleurs and St. Augustine, and 256 kbps/192 kbps between STUD3 and Champs Fleurs. ACLs were applied to control traffic between Champs Fleurs and STUD4.

Issues Encountered

Issue	Resolution
One group member (Allin) became unresponsive during implementation	The remaining members redistributed his tasks and configured STUD3 themselves
Frame Relay DLCI mapping confusion	Created a detailed mapping table before configuration
GNS3 interface limitations (only 2 FastEthernet ports)	Used switches and subinterfaces to connect multiple STUD routers
IP addressing inconsistencies between the plan and the implementation	Updated IP table to match actual running configs

Limitations

- GNS3 simulation does not perfectly replicate real hardware behaviour
- Frame Relay CIR enforcement is simulated and may not reflect carrier-grade implementation
- Router interface count limited available connections

8 Conclusion

This project successfully demonstrated the design and implementation of a complex WAN infrastructure incorporating multiple networking technologies, including Frame Relay, eBGP, GRE VPN, NAT, DHCP, NTP, DNS, and ACLs. The Group 3 (DataKrash) team effectively integrated these technologies into a cohesive network topology simulated in GNS3, meeting all specified requirements.

The project was highly successful, with all primary objectives achieved: complete traceroute documentation from three of four group members, a functional Frame Relay network with CIR limitations, full eBGP configuration with route propagation across all seven routers, an operational GRE VPN tunnel between STUD1 and STUD3, working NAT/PAT on STUD3 and STUD4, operational DHCP, NTP, and DNS services, and effective ACL implementation. End-to-end connectivity was verified through comprehensive testing.

JASMIN HIPPOLYTE
816036300
INFO 3605 – Fundamentals of LAN Technologies

MOHAMMED IQBAL
816041782
INFO 3605 – Fundamentals of LAN Technologies

RENUAH SAMMUEL
816040963
INFO 3605 – Fundamentals of LAN Technologies

Recommendations for Future Project

Security

- Add IPsec encryption to GRE tunnels
- Use BGP MD5 authentication between peers
- Implement port security on access switches

Redundancy

- Add backup Frame Relay PVCs
- Configure HSRP for gateway failover
- Use BGP multipath for load balancing

Automation

- Use Ansible or Python scripts to deploy configs
- Automate configuration backups

Modern Technologies

- Replace Frame Relay with MPLS or SD-WAN
- Implement IPv6 dual-stack

Monitoring

- Add SNMP monitoring
- Configure syslog server for centralised logging

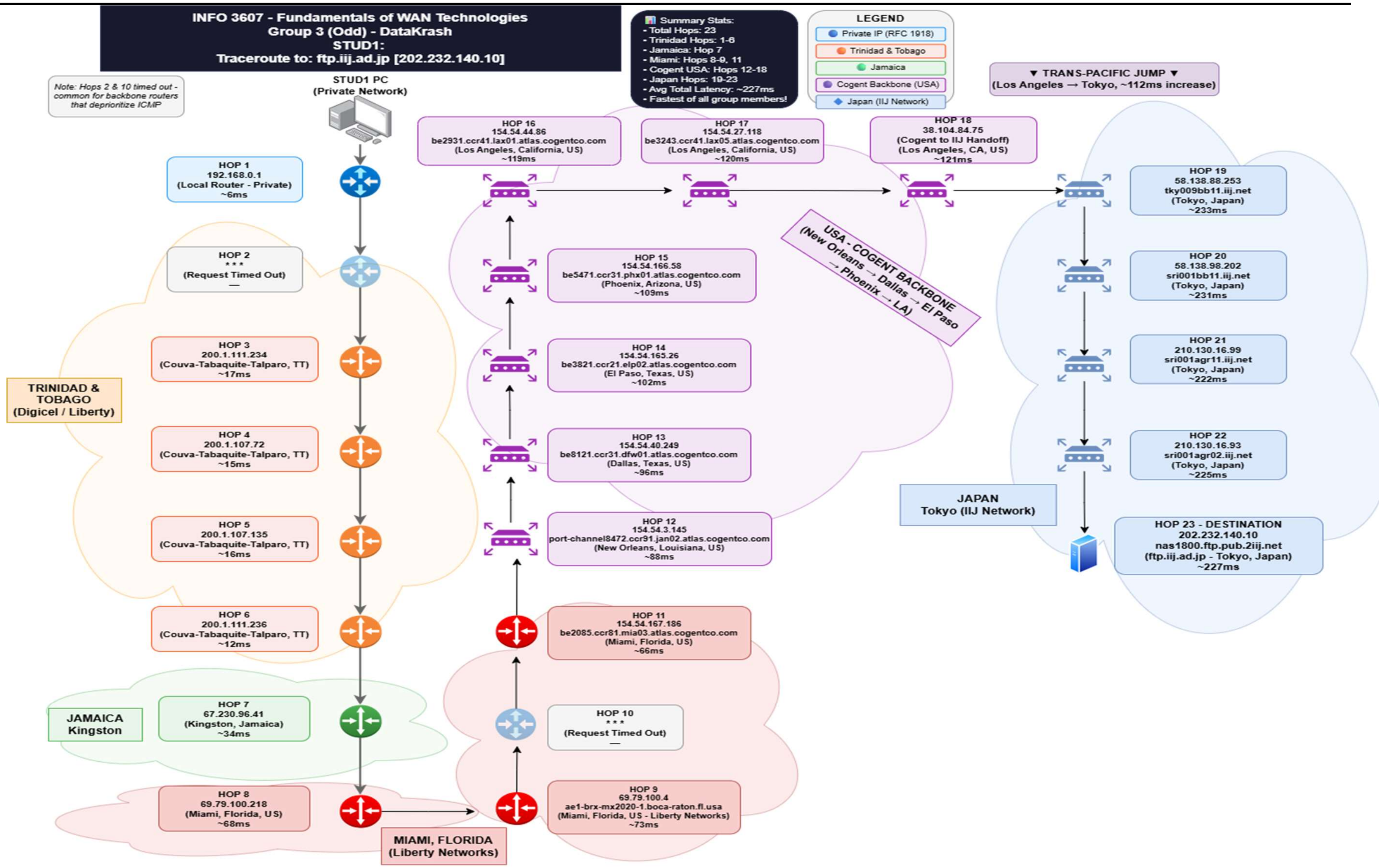
Testing

- Create formal test plans before implementation
- Peer review all configs before applying

Documentation

9 Maintain real-time troubleshooting logs
10 Include network diagrams with interface labels

11 Appendix I - Output



Group Number: 3 (Odd Group)
Target IP Address: 202.232.140.10 (ftp.iij.ad.jp)

STUD1 (816036300) Jasmin Hippolyte:

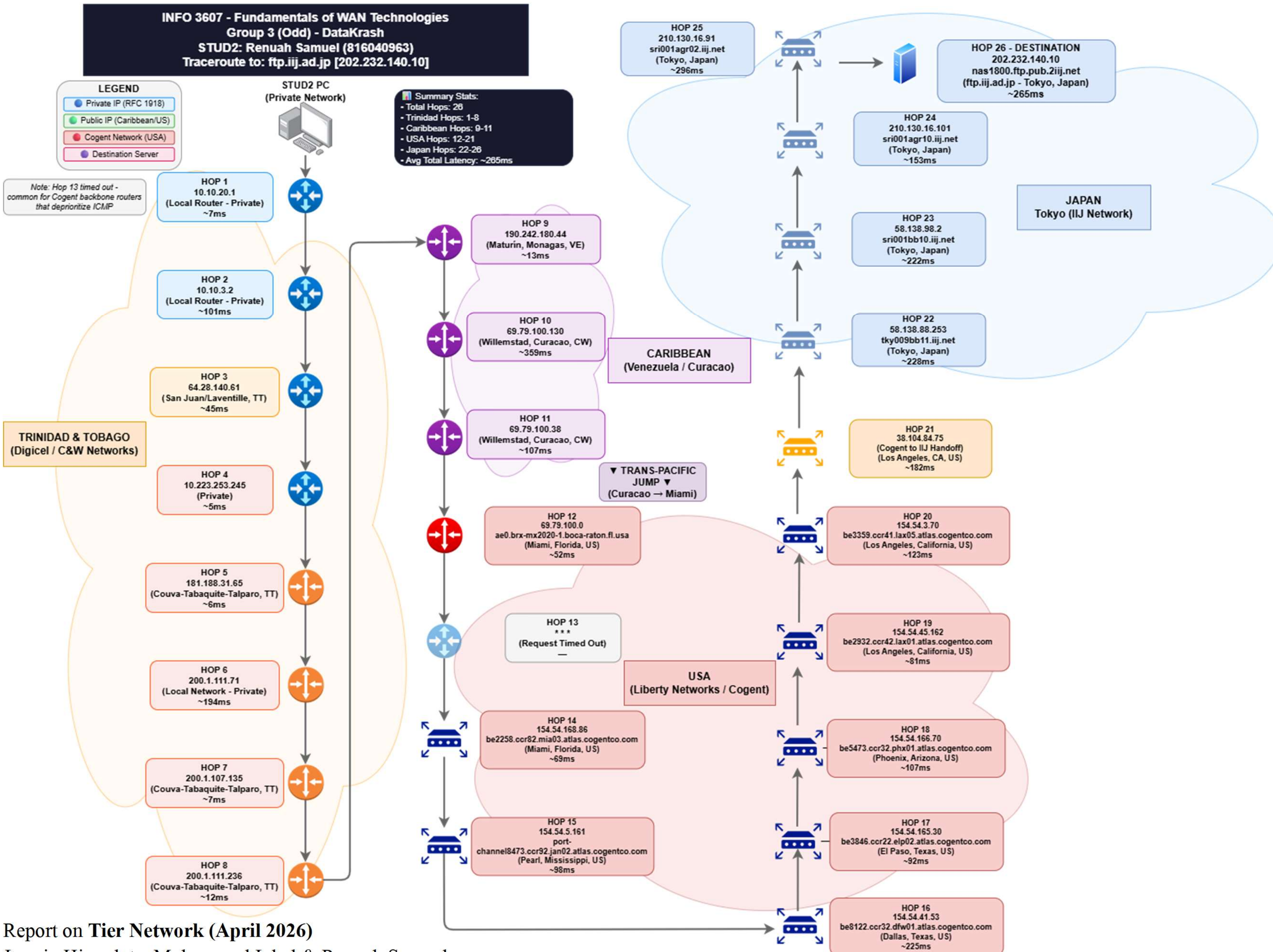
<u>Traceroute Result</u>

```
C:\Users\Jasmin Hippolyte>tracert 202.232.140.10

Tracing route to nas1800.ftp.pub.2iij.net [202.232.140.10]
over a maximum of 30 hops:

  1     9 ms     5 ms     3 ms  192.168.0.1
  2     *        *        *     Request timed out.
  3    20 ms    13 ms    19 ms  200.1.111.234
  4    20 ms    13 ms    12 ms  200.1.107.72
  5    20 ms    14 ms    15 ms  200.1.107.135
  6    12 ms    13 ms    11 ms  200.1.111.236
  7    37 ms    32 ms    33 ms  67.230.96.41
  8    72 ms    68 ms    65 ms  69.79.100.218
  9    74 ms    74 ms    70 ms  ae1-brx-mx2020-1.boca-raton.fl.usa.libertynetworks.com [69.79.100.4]
 10     *        *        *     Request timed out.
 11    66 ms     *        *     be2085.ccr81.mia03.atlas.cogentco.com [154.54.167.186]
 12    89 ms    90 ms    86 ms  port-channel8472.ccr91.jan02.atlas.cogentco.com [154.54.3.145]
 13   111 ms    89 ms    88 ms  be8121.ccr31.dfw01.atlas.cogentco.com [154.54.40.249]
 14   100 ms   103 ms   102 ms  be3821.ccr21.elp02.atlas.cogentco.com [154.54.165.26]
 15   109 ms   109 ms   110 ms  be5471.ccr31.phx01.atlas.cogentco.com [154.54.166.58]
 16   119 ms   120 ms   119 ms  be2931.ccr41.lax01.atlas.cogentco.com [154.54.44.86]
 17   120 ms   121 ms   120 ms  be3243.ccr41.lax05.atlas.cogentco.com [154.54.27.118]
 18   128 ms   117 ms   119 ms  38.104.84.75
 19   238 ms   232 ms   230 ms  tky009bb11.IIJ.Net [58.138.88.253]
 20   232 ms   230 ms   232 ms  sri001bb11.IIJ.Net [58.138.98.202]
 21   222 ms   221 ms   222 ms  sri001agr11.iij.net [210.130.16.99]
 22   226 ms   225 ms   225 ms  sri001agr02.iij.net [210.130.16.93]
 23   227 ms   227 ms   226 ms  nas1800.ftp.pub.2iij.net [202.232.140.10]

Trace complete.
```

Hop Table STUD1

Hop #	IP Address	Hostname	Location	Avg Latency (ms)
1	192.168.0.1	(Private)	Local Network	~6ms
2	*	Request Timed Out	*	*
3	200.1.111.234	(Private)	Couva-Tabaquite-Talparo, TT	~17ms
4	200.1.107.72	(Private)	Couva-Tabaquite-Talparo, TT	15ms
5	200.1.107.135	(Private)	Couva-Tabaquite-Talparo, TT	~16ms
6	200.1.111.236	(Private)	Couva-Tabaquite-Talparo, TT	12ms
7	67.230.96.41	–	Kingston, Jamaica	34ms
8	69.79.100.218	–	Miami, Florida, US	~68ms
9	69.79.100.4	ae1-brx-mx2020-1.boca-raton.fl.usa.libertynetworks.com	Miami, Florida, US	~73ms
10	*	Request Timed Out	*	*
11	154.54.167.186	be2085.ccr81.mia03.atlas.cogentco.com	Miami, Florida, US	66ms
12	154.54.3.145	port-channel8472.ccr91.jan02.atlas.cogentco.com	New Orleans, Louisiana, US	~88ms

13	154.54.40.249	be8121.ccr31.dfw01.atlas.cogentco.com	Dallas, Texas, US	96ms
14	154.54.165.26	be3821.ccr21.elp02.atlas.cogentco.com	El Paso, Texas, US	~102ms
15	154.54.166.58	be5471.ccr31.phx01.atlas.cogentco.com	Phoenix, Arizona, US	~109ms
16	154.54.44.86	be2931.ccr41.lax01.atlas.cogentco.com	Los Angeles, California, US	~119ms
17	154.54.27.118	be3243.ccr41.lax05.atlas.cogentco.com	Los Angeles, California, US	~120ms
18	38.104.84.75	–	Los Angeles, California, US	~121ms
19	58.138.88.253	tky009bb11.iij.net	Tokyo, Japan	~233ms
20	58.138.98.202	sri001bb11.iij.net	Tokyo, Japan	~231ms
21	210.130.16.99	sri001agr11.iij.net	Tokyo, Japan	~222ms
22	210.130.16.93	sri001agr02.iij.net	Tokyo, Japan	~225ms
23	**202.232.140.10**	**nas1800.ftp.pub.2iij.net**	**Tokyo, Japan**	**~227ms**


INFO 3607 - Fundamentals of WAN Technologies
Group 3 (Odd) - DataKrash
STUD2: Renuah Samuel (816040963)
Traceroute to: ftp.iij.ad.jp [202.232.140.10]
LEGEND
Private IP (RFC 1918)
Public IP (Caribbean/US)
Cogent Network (USA)
Destination Server
STUD2 PC
(Private Network)
Summary Stats:
• Total Hops: 26
• Trinidad Hops: 1-8
• Caribbean Hops: 9-11
• USA Hops: 12-21
• Japan Hops: 22-26
• Avg Total Latency: ~265ms
Note: Hop 13 timed out - common for Cogent backbone routers that deprioritize ICMP
HOP 1 10.10.20.1 (Local Router - Private) ~7ms
HOP 2 10.10.3.2 (Local Router - Private) ~101ms
HOP 3 64.28.140.61 (San Juan/Laventille, TT) ~45ms
TRINIDAD & TOBAGO (Digicel / C&W Networks)
HOP 4 10.223.253.245 (Private) ~5ms
HOP 5 181.188.31.65 (Couva-Tabaquite-Talparo, TT) ~6ms
HOP 6 200.1.111.71 (Local Network - Private) ~194ms
HOP 7 200.1.107.135 (Couva-Tabaquite-Talparo, TT) ~7ms
HOP 8 200.1.111.236 (Couva-Tabaquite-Talparo, TT) ~12ms
HOP 9 190.242.180.44 (Maturin, Monagas, VE) ~13ms
HOP 10 69.79.100.130 (Willemstad, Curacao, CW) ~359ms
CARIBBEAN (Venezuela / Curacao)
HOP 11 69.79.100.38 (Willemstad, Curacao, CW) ~107ms
▼ TRANS-PACIFIC JUMP ▼ (Curacao → Miami)
HOP 12 69.79.100.0 ae0.brx-mx2020-1.boca-raton.fl.usa (Miami, Florida, US) ~52ms
HOP 13 * * * (Request Timed Out) —
USA (Liberty Networks / Cogent)
HOP 14 154.54.168.86 be2258.ccr82.mia03.atlas.cogentco.com (Miami, Florida, US) ~69ms
HOP 15 154.54.5.161 port-channel8473.ccr92.jan02.atlas.cogentco.com (Pearl, Mississippi, US) ~98ms
HOP 16 154.54.41.53 be8122.ccr32.dfw01.atlas.cogentco.com (Dallas, Texas, US) ~225ms
HOP 17 154.54.165.30 be3846.ccr22.elp02.atlas.cogentco.com (El Paso, Texas, US) ~92ms
HOP 18 154.54.166.70 be5473.ccr32.phx01.atlas.cogentco.com (Phoenix, Arizona, US) ~107ms
HOP 19 154.54.45.162 be2932.ccr42.lax01.atlas.cogentco.com (Los Angeles, California, US) ~81ms
HOP 20 154.54.3.70 be3359.ccr41.lax05.atlas.cogentco.com (Los Angeles, California, US) ~123ms
HOP 21 38.104.84.75 (Cogent to IIJ Handoff) (Los Angeles, CA, US) ~182ms
HOP 22 58.138.88.253 tky009bb11.iij.net (Tokyo, Japan) ~228ms
HOP 23 58.138.98.2 sri001bb10.iij.net (Tokyo, Japan) ~222ms
JAPAN Tokyo (IIJ Network)
HOP 24 210.130.16.101 sri001agr10.iij.net (Tokyo, Japan) ~153ms
HOP 25 210.130.16.91 sri001agr02.iij.net (Tokyo, Japan) ~296ms
HOP 26 - DESTINATION 202.232.140.10 nas1800.ftp.pub.2iij.net (ftp.iij.ad.jp - Tokyo, Japan) ~265ms

STUD2 (816040963) Renuah Samuel:

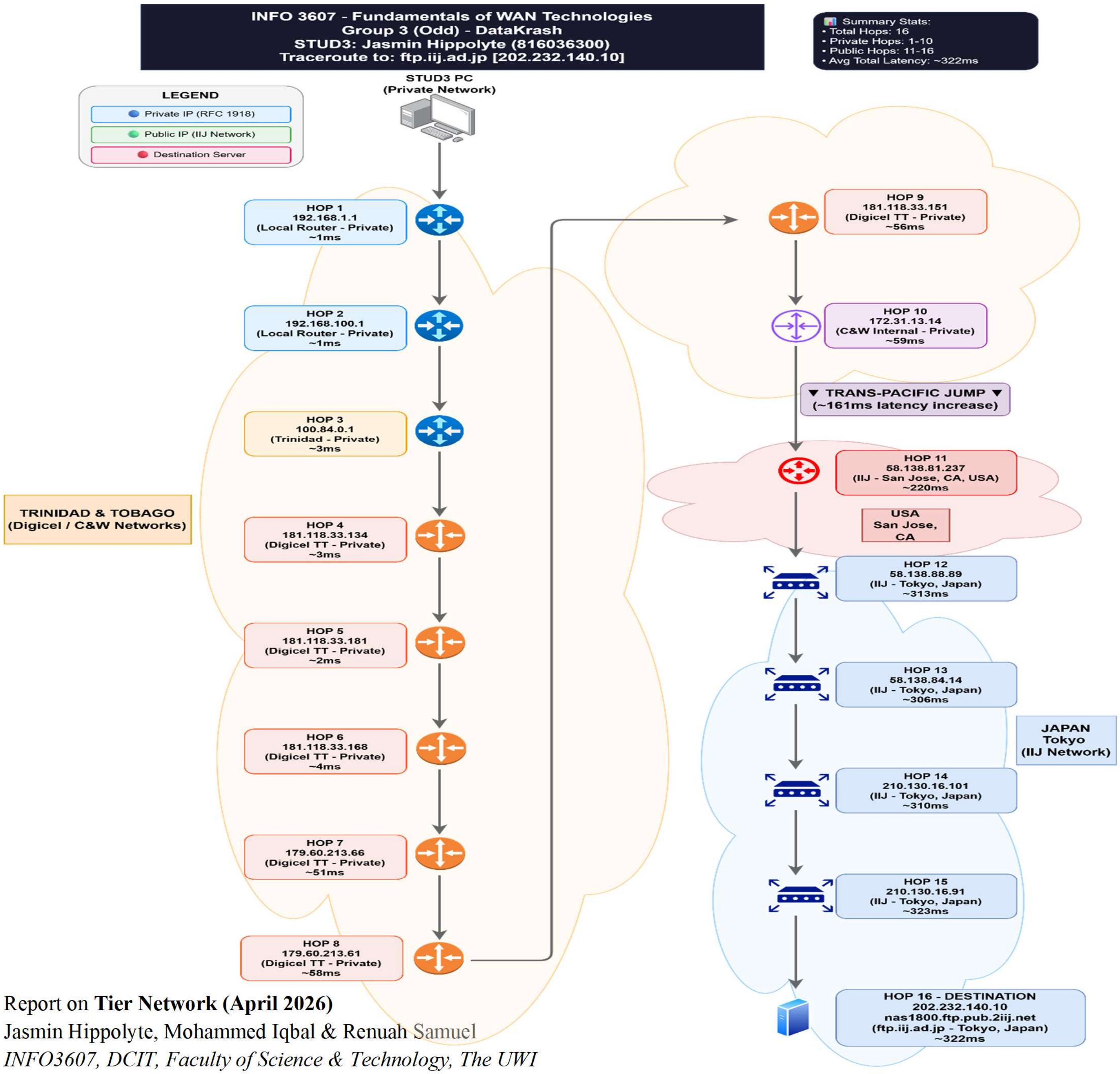
<u>Traceroute Result STUD2</u>

```
Command Prompt

Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Renuah Samuel>tracert 202.232.140.10

Tracing route to nas1800.ftp.pub.2iij.net [202.232.140.10]
over a maximum of 30 hops:

  1     8 ms     8 ms     4 ms  10.10.20.1
  2   211 ms    68 ms    24 ms  10.10.3.2
  3    95 ms     5 ms    34 ms  64.28.140.61
  4     9 ms     3 ms     4 ms  10.223.253.245
  5     6 ms     5 ms     6 ms  181.188.31.65
  6   460 ms   100 ms    23 ms  200.1.111.71
  7     5 ms    10 ms     6 ms  200.1.107.135
  8     6 ms     5 ms    24 ms  200.1.111.236
  9    22 ms     6 ms    11 ms  190.242.180.44
 10   207 ms   213 ms   656 ms  69.79.100.130
 11   203 ms    65 ms    54 ms  69.79.100.38
 12    52 ms    53 ms    52 ms  ae0.brx-mx2020-1.boca-raton.fl.usa.libertynetworks.com [69.79.100.0]
 13     *        *        *     Request timed out.
 14     *        *       69 ms  be2258.ccr82.mia03.atlas.cogentco.com [154.54.168.86]
 15   136 ms    81 ms    78 ms  port-channel8473.ccr92.jan02.atlas.cogentco.com [154.54.5.161]
 16    79 ms   123 ms   473 ms  be8122.ccr32.dfw01.atlas.cogentco.com [154.54.41.53]
 17    95 ms    91 ms    91 ms  be3846.ccr22.elp02.atlas.cogentco.com [154.54.165.30]
 18   101 ms   105 ms   116 ms  be5473.ccr32.phx01.atlas.cogentco.com [154.54.166.70]
 19   130 ms   114 ms     *     be2932.ccr42.lax01.atlas.cogentco.com [154.54.45.162]
 20   112 ms   129 ms   128 ms  be3359.ccr41.lax05.atlas.cogentco.com [154.54.3.70]
 21   311 ms   119 ms   115 ms  38.104.84.75
 22   250 ms   217 ms   218 ms  tky009bb11.iij.net [58.138.88.253]
 23   216 ms   227 ms   222 ms  sri001bb10.iij.net [58.138.98.2]
 24     *      230 ms   228 ms  sri001agr10.iij.net [210.130.16.101]
 25   231 ms   410 ms   248 ms  sri001agr02.iij.net [210.130.16.91]
 26   359 ms   219 ms   216 ms  nas1800.ftp.pub.2iij.net [202.232.140.10]

Trace complete.

C:\Users\Renuah Samuel>
```

Hop Table STUD2

Hop #	IP Address	Hostname	Location	Avg Latency (ms)
1	10.10.20.1	(Private)	Local Network	~7ms
2	10.10.3.2	(Private)	Local Network	~101ms
3	64.28.140.61	–	San Juan/Laventille, TT	~45ms
4	10.223.253.245	(Private)	(Private)	~5ms
5	181.188.31.65	(Private)	Couva-Tabaquite-Talparo, TT	~6ms
6	200.1.111.71	(Private)	Local Network	~194ms
7	200.1.107.135	(Private)	Couva-Tabaquite-Talparo, TT	~7ms
8	200.1.111.236	(Private)	Couva-Tabaquite-Talparo, TT	~12ms
9	190.242.180.44	(Private)	Maturín, Monagas, VE	~13ms
10	69.79.100.130	(Private)	Willemstad, Curacao, CW	~359ms
11	69.79.100.38	(Private)	Willemstad, Curacao, CW	~107ms
12	69.79.100.0	ae0.brx-mx2020-1.boca-raton.fl.usa.libertynetworks.com	Miami, Florida, US	~52ms
13	*	Request timed out	–	–
14	154.54.168.86	be2258.ccr82.mia03.atlas.cogentco.com	Miami, Florida, US	~69ms
15	154.54.5.161	port-	Pearl, Mississippi, US	~98ms

Hop #	IP Address	Hostname	Location	Avg Latency (ms)
		channel8473.ccr92.jan02.atlas.cogentco.com		
16	154.54.41.53	be8122.ccr32.dfw01.atlas.cogentco.com	Dallas, Texas, US	~225ms
17	154.54.165.30	be3846.ccr22.elp02.atlas.cogentco.com	El Paso, Texas, US	~92ms
18	154.54.166.70	be5473.ccr32.phx01.atlas.cogentco.com	Phoenix, Arizona, US	~107ms
19	154.54.45.162	be2932.ccr42.lax01.atlas.cogentco.com	Los Angeles, California, US	~81ms
20	154.54.3.70	be3359.ccr41.lax05.atlas.cogentco.com	Los Angeles, California, US	~123ms
21	38.104.84.75	(Cogent to IIJ handoff)	Los Angeles, California, US	~182ms
22	58.138.88.253	tky009bb11.iij.net	Tokyo, Japan	~228ms
23	58.138.98.2	sri001bb10.iij.net	Tokyo, Japan	~222ms
24	210.130.16.101	sri001agr10.iij.net	Tokyo, Japan	~153ms
25	210.130.16.91	sri001agr02.iij.net	Tokyo, Japan	~296ms
26	**202.232.140.10**	**nas1800.ftp.pub.2iij.net**	**Tokyo, Japan**	**~265ms**



Report on **Tier Network (April 2026)**
Jasmin Hippolyte, Mohammed Iqbal & Renuah Samuel
INFO3607, DCIT, Faculty of Science & Technology, The UWI

STUD3 (816036300) Jasmin Hippolyte:

<u>Traceroute Result STUD3</u>

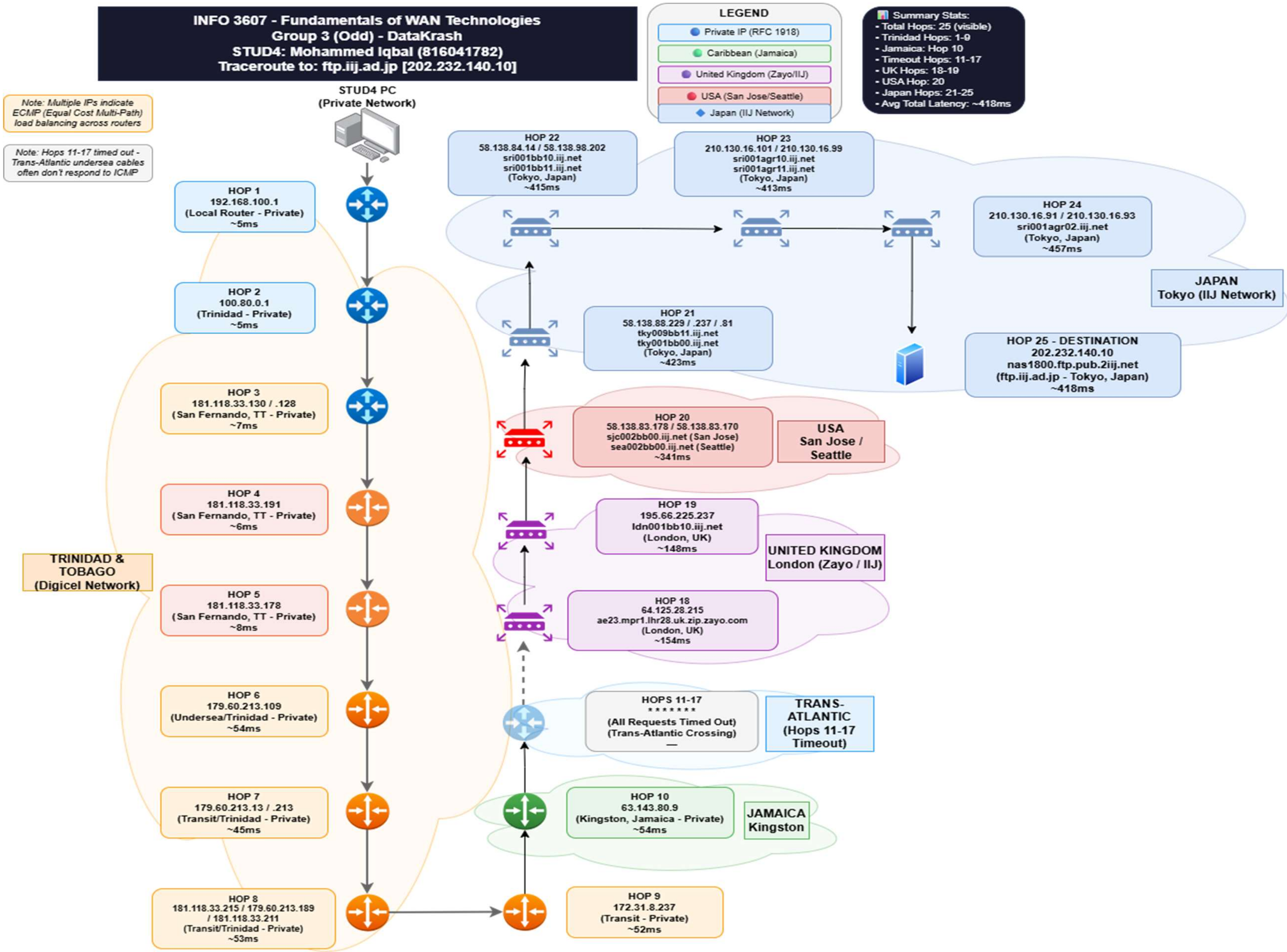
Command Prompt

```
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Bye Pass NTCOG>tracert 202.232.140.10

Tracing route to nas1800.ftp.pub.2iij.net [202.232.140.10]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.1.1
  2    <1 ms    <1 ms    <1 ms  192.168.100.1
  3     4 ms     2 ms     2 ms  100.84.0.1
  4     4 ms     3 ms     3 ms  181.118.33.134
  5     2 ms     2 ms     2 ms  181.118.33.181
  6     4 ms     4 ms     4 ms  181.118.33.168
  7    52 ms    51 ms    51 ms  179.60.213.66
  8    58 ms    58 ms    58 ms  179.60.213.61
  9    56 ms    56 ms    56 ms  181.118.33.151
 10    60 ms    59 ms    59 ms  172.31.13.14
 11   220 ms   220 ms   220 ms  sjc002bb01.IIJ.Net [58.138.81.237]
 12   313 ms   312 ms   313 ms  tky001bb00.IIJ.Net [58.138.88.89]
 13   306 ms   306 ms   306 ms  sri001bb10.IIJ.Net [58.138.84.14]
 14   311 ms   310 ms   310 ms  sri001agr10.iij.net [210.130.16.101]
 15   323 ms   323 ms   322 ms  sri001agr02.iij.net [210.130.16.91]
 16   322 ms   322 ms   322 ms  nas1800.ftp.pub.2iij.net [202.232.140.10]

Trace complete.

C:\Users\Bye Pass NTCOG>
```

Hop Table STUD3

Hop #	IP Address	Hostname	Location	Avg Latency (ms)
1	192.168.1.1	(Private)	Local Network	~1ms
2	192.168.100.1	(Private)	Local Network	~1ms
3	100.84.0.1	(Private)	Trinidad	~3ms
4	181.118.33.134	(Private)	Digicel, Trinidad & Tobago	~3ms
5	181.118.33.181	(Private)	Digicel, Trinidad & Tobago	~2ms
6	181.118.33.168	(Private)	Digicel, Trinidad & Tobago	~4ms
7	179.60.213.66	(Private)	Digicel, Trinidad & Tobago	~51ms
8	179.60.213.61	(Private)	Digicel, Trinidad & Tobago	~58ms
9	181.118.33.151	(Private)	Digicel, Trinidad & Tobago	~56ms
10	172.31.13.14	(Private)	C&W Internal (private)	~59ms
11	58.138.81.237	sjc002bb01.iij.net	San Jose, California / Internet Initiative Japan	~220ms
12	58.138.88.89	tky001bb00.iij.net	Tokyo, Japan / Internet Initiative Japan	~313ms
13	58.138.84.14	sri001bb10.iij.net	Tokyo, Japan / Internet Initiative Japan	~306ms
14	210.130.16.101	sri001agr10.iij.net	Tokyo, Japan / Internet Initiative Japan	~310ms

15	210.130.16.91	sri001agr02.iij.net	Tokyo, Japan / Internet Initiative Japan	~323ms
16	**202.232.140.10**	**nas1800.ftp.pub.2iij.net**	**Tokyo, Japan / Internet Initiative Japan**	**~322ms**



Report on **Tier Network (April 2026)**
Jasmin Hippolyte, Mohammed Iqbal & Renuah Samuel
INFO3607, DCIT, Faculty of Science & Technology, The UWI

STUD4 (816041782) Mohammed Iqbal:

Traceroute Result STUD4

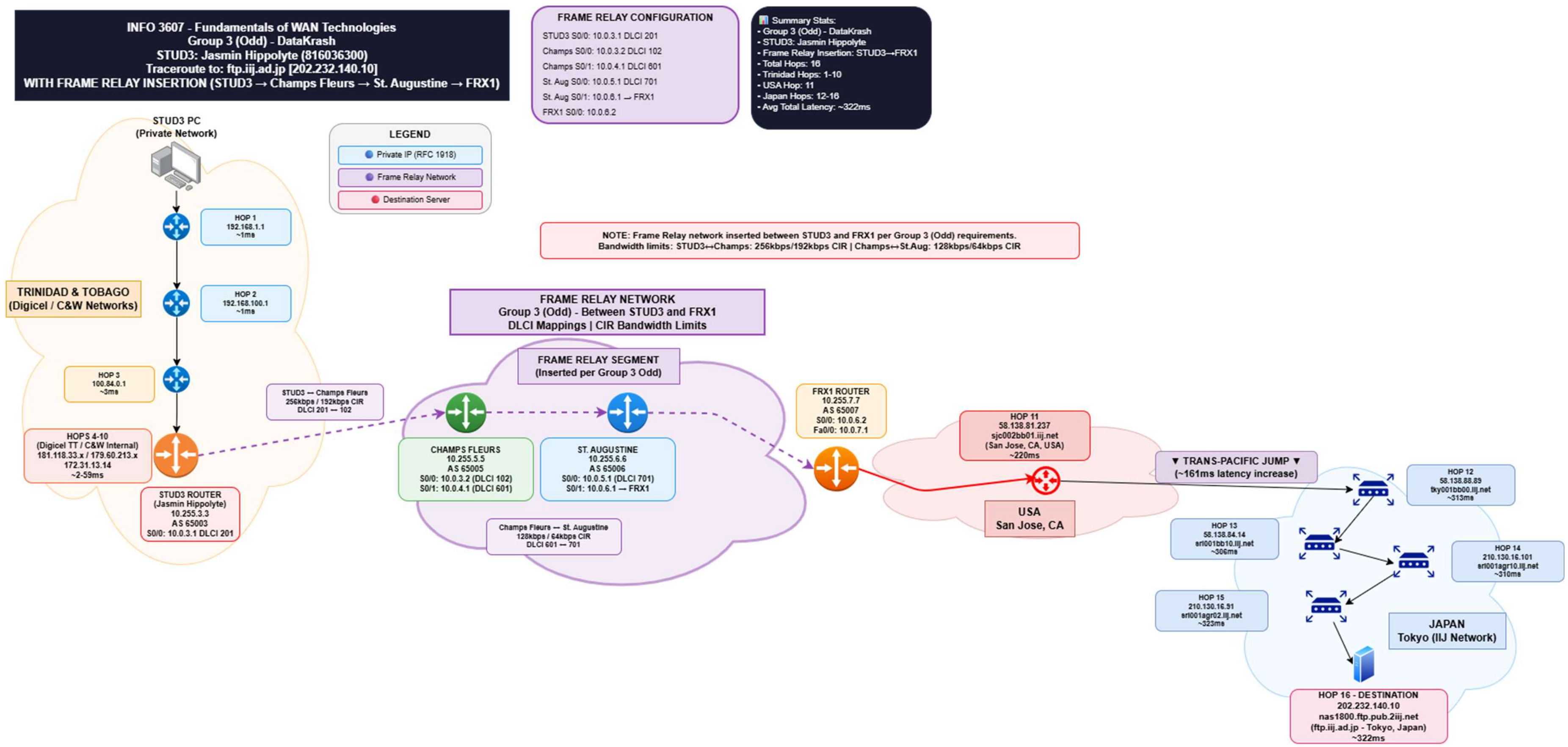
```
a--— -zsh — 189×41
a--@s-MacBook-Air ~ % traceroute 202.232.140.10
traceroute to 202.232.140.10 (202.232.140.10), 64 hops max, 40 byte packets
 1  192.168.100.1 (192.168.100.1)  9.265 ms  2.981 ms  2.852 ms
 2  100.80.0.1 (100.80.0.1)  5.582 ms  5.593 ms  5.538 ms
 3  181.118.33.128 (181.118.33.128)  9.490 ms
    181.118.33.130 (181.118.33.130)  7.149 ms
    181.118.33.128 (181.118.33.128)  7.471 ms
 4  181.118.33.191 (181.118.33.191)  6.754 ms  6.470 ms  6.080 ms
 5  181.118.33.178 (181.118.33.178)  8.771 ms  7.886 ms  8.685 ms
 6  179.60.213.109 (179.60.213.109)  53.288 ms  54.099 ms  54.012 ms
 7  179.60.213.13 (179.60.213.13)  47.649 ms
    179.60.213.213 (179.60.213.213)  43.715 ms
    179.60.213.13 (179.60.213.13)  44.225 ms
 8  181.118.33.6 (181.118.33.6)  51.011 ms
    181.118.33.211 (181.118.33.211)  51.242 ms  50.600 ms
 9  172.31.8.237 (172.31.8.237)  51.525 ms  49.215 ms  52.546 ms
10  63.143.80.9 (63.143.80.9)  49.386 ms  54.761 ms  50.158 ms
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  ae23.mpr1.lhr28.uk.zip.zayo.com (64.125.28.215)  162.997 ms  160.222 ms  154.745 ms
19  ldn001bb10.iij.net (195.66.225.237)  154.423 ms  151.716 ms  148.781 ms
20  sjc002bb00.iij.net (58.138.83.178)  413.833 ms  406.656 ms  309.744 ms
21  tky009bb11.iij.net (58.138.88.237)  447.297 ms
    tky001bb00.iij.net (58.138.88.129)  506.643 ms  509.565 ms
22  sri001bb10.iij.net (58.138.84.14)  512.371 ms
    sri001bb11.iij.net (58.138.98.202)  510.981 ms
    sri001bb10.iij.net (58.138.84.14)  511.126 ms
23  sri001agr10.iij.net (210.130.16.101)  508.400 ms
    sri001agr11.iij.net (210.130.16.99)  509.534 ms
    sri001agr10.iij.net (210.130.16.101)  510.818 ms
24  sri001agr02.iij.net (210.130.16.91)  509.275 ms  511.576 ms  511.023 ms
25  nas1800.ftp.pub.2iij.net (202.232.140.10)  512.080 ms  421.421 ms  497.182 ms
a--@s-MacBook-Air ~ %
```

Hop Table STUD4

Hop #	IP Address	Hostname	Location	Avg Latency
1	192.168.100.1	(Private)	Trinidad	~5ms
2	100.80.0.1	(Private)	Trinidad	~5ms
3	181.118.33.130 and 181.118.33.128	(Private)	San Fernando, Trinidad	~7ms
4	181.118.33.191	(Private)	San Fernando, Trinidad	~6ms
5	181.118.33.178	(Private)	San Fernando, Trinidad	~8ms
6	179.60.213.109	(Private)	Undersea/Trinidad	~54ms
7	179.60.213.13 and 179.60.213.213	(Private)	Transit/Trinidad	~45ms
8	181.118.33.215, 179.60.213.189 and 181.118.33.211	(Private)	Transit/Trinidad	~53ms
9	172.31.8.237	(Private)	Transit	~52ms
10	63.143.80.9	(Private)	Kingston, Jamaica	~54ms
11–17	* * *	—	Unknown	—

18	64.125.28.215	ae23.mpr1.lhr28.uk.zip.zayo.com	London, UK	~154ms
19	195.66.225.237	ldn001bb10.iij.net	London, UK	~148ms
20	58.138.83.178 and 58.138.83.170	sjc002bb00.iij.net and sea002bb00.iij.net	San Jose, USA	~341ms
21	58.138.88.229, 58.138.88.237 and 58.138.88.81	tky009bb11.iij.net and tky001bb00.iij.net	Tokyo, Japan	~423ms
22	58.138.84.14 and 58.138.98.202	sri001bb10.iij.net and sri001bb11.iij.net	Tokyo, Japan	~415ms
23	210.130.16.101 and 210.130.16.99	sri001agr10.iij.net and sri001agr11.iij.net	Tokyo, Japan	~413ms
24	210.130.16.91 and 210.130.16.93	sri001agr02.iij.net	Tokyo, Japan	~457ms
25	**202.232.140.10**	**nas1800.ftp.pub.2iij.net**	**Tokyo, Japan**	**~418ms**

STUD3 WITH FRAME RELAY DIAGRAM (Updated)



Report on **Tier Network (April 2026)**
Jasmin Hippolyte, Mohammed Iqbal & Renuah Samuel
INFO3607, DCIT, Faculty of Science & Technology, The UWI

eBGP Layout

Router	AS Number
STUD1 (Jasmin)	65001
STUD2 (Renuah)	65002
STUD3	65003
STUD4 (Mohammed)	65004
Champ Fleurs	65005
St. Augustine	65006
FRX1	65007
FTP-Server	65008

Network Links and Bandwidth Table

SiteA	SiteB	Link Bandwidth	CIR	Comments
All Groups				
Champ Fleurs	St. Augustine	128kbps	64kbps	
Group Number Odd				
STUD3 ID NUM	Champ Fleurs	256kbps	192kbps	

ACL (Access Control Lists)

SiteA	SiteB	Allow	Deny	Comments
Group No. ODD				
STUD2	STUD4	HTTP/HTTPS, eBGP, CMP, DHCP, NTP, DNS, SSH	Any	Applied on the GRE VPN Tunnel
All Groups				
Champ Fleurs	STUD4	FTP, eBGP, ICMP, DHCP, NTP, DNS, SSH	Any	

IP Addressing and Networks

Organization Rules

Type	Network	Purpose
Frame Relay Core	10.1.1.0 /24	FR cloud connections (STUD3, Champs, St. Aug, FRX1)
STUD Links	10.10.x.0 /30	STUD1, STUD2, STUD4 to FRX1
GRE Tunnel	172.16.1.0 /30	STUD1 ↔ STUD3 VPN
LANs (PCs/Devices)	192.168.x.0 /24	Client PCs
Destination	202.232.140.0 /24	FTPServer

GRE Tunnel (172.16.X.X)

Tunnel	Network	STUD1 IP	STUD3 IP
STUD1 ↔ STUD3 (VPN)	172.16.1.0/30	172.16.1.1	172.16.1.2

LAN Networks (192.168.X.X) - For Client PCs

Location	Network	Gateway (Router IP)	DHCP Range
STUD1 LAN	192.168.11.0 /24	192.168.11.1	192.168.11.100 - 200
STUD2 LAN	192.168.2.0 /24	192.168.2.1	192.168.2.100 - 200

STUD3 LAN	192.168.3.0 /24	192.168.3.1	192.168.3.100 - 200
STUD4 LAN	192.168.4.0 /24	192.168.4.1	192.168.4.100 - 200
StAugustine LAN	192.168.1.10 /24	192.168.1.1	192.168.1.100 - 200
ChampsFleurs LAN	192.168.5.10 /24	192.168.5.1	192.168.5.100 - 200

Main IP Addressing Table

Name	Interface	IPV4 Address	Subnet Mask	Network Address	Broadcast Address	Gateway	Comment
Champ Fleurs							
	S0/0	10.1.5.2	255.255.255.252	10.1.5.0	10.1.5.3	10.1.5.1	Link to FR3 (Frame Relay)
	Fa1/0	192.168.5.1	255.255.255.0	192.168.5.0	192.168.5.255	N/A	Champs Fleurs LAN Gateway
	Loopback0	10.255.5.1	255.255.255.255	10.255.5.1	10.255.5.1	N/A	Router ID for eBGP
St. Augustine							
	S0/0	10.1.6.2	255.255.255.252	10.1.6.0	10.1.6.3	N/A	Link to FR4 (Frame Relay)
	Fa1/0	192.168.6.1	255.255.255.0	192.168.6.0	192.168.6.255	N/A	St. Augustine LAN Gateway
	Loopback0	10.255.6.1	255.255.255.255	10.255.6.1	10.255.6.1	N/A	Router ID for eBGP

FRX1							
	S0/0	10.1.8.2	255.255.255.252	10.1.8.0	10.1.8.3	N/A	Link to FR2 (Frame Relay)
	S0/1	10.0.1.2	255.255.255.252	10.0.1.0	10.0.1.3	10.0.1.1	Link to STUD1
	S0/2	10.0.2.2	255.255.255.252	10.0.2.0	10.0.2.3	10.0.2.1	Link to STUD2
	S0/3	10.0.4.2	255.255.255.252	10.0.4.0	10.0.4.3	10.0.4.1	Link to STUD4
	Loopback0	10.255.7.1	255.255.255.255	10.255.7.1	10.255.7.1	N/A	Router ID for eBGP
FR1							
	S0/1	10.1.2.1	255.255.255.252	10.1.2.0	10.1.2.3	N/A	Link to FR2
	S0/2	10.1.3.1	255.255.255.252	10.1.3.0	10.1.3.3	N/A	Link to FR3
	S0/3	10.1.1.1	255.255.255.252	10.1.1.0	10.1.1.3	N/A	Link to STUD3
	Loopback0	10.255.11.1	255.255.255.255	10.255.11.1.	10.255.11.1	N/A	Management
FR2							
	S0/0	10.1.8.1	255.255.255.252	10.1.8.0	10.1.8.3	N/A	Link to FRX1
	S0/1	10.1.2.2	255.255.255.252	10.1.2.0	10.1.2.3	N/A	Link to FR1
	S0/2	10.1.7.2	255.255.255.252	10.1.7.0	10.1.7.3	N/A	Link to FR4
	Loopback0	10.255.12.1	255.255.255.255	10.255.12.1.	10.255.12.1	N/A	Management

FR3							
	S0/0	10.1.5.1	255.255.255.252	10.1.5.0	10.1.5.3	N/A	Link to Champs Fleurs
	S0/1	10.1.3.2	255.255.255.252	10.1.3.0	10.1.3.3	N/A	Link to FR1
	S0/2	10.1.4.1	255.255.255.252	10.1.4.0	10.1.4.3	N/A	Link to FR4
	Loopback0	10.255.13.1	255.255.255.255	10.255.13.1.	10.255.13.1	N/A	Management
FR4							
	S0/0	10.1.6.1	255.255.255.252	10.1.6.0	10.1.6.3	N/A	Link to St. Augustine
	S0/1	10.1.7.1	255.255.255.252	10.1.7.0	10.1.7.3	N/A	Link to FR2
	S0/2	10.1.4.2	255.255.255.252	10.1.4.0	10.1.4.3	N/A	Link to FR3
	Loopback0	10.255.14.1	255.255.255.255	10.255.14.1.	10.255.14.1	N/A	Management
FTP-Server							
	Fa0/1	10.101.22.2	255.255.255.252	10.101.22.0	10.101.22.3	10.101.22.1	S1-HOP22 Fa0/0 (10.101.22.1)
	Fa2/0	10.102.25.2	255.255.255.252	10.102.25.0	10.102.25.3	10.102.25.1	S2-HOP25 Fa0/0 (10.102.25.1)
	Fa3/0	10.103.15.2	255.255.255.252	10.103.15.0	10.103.15.3	10.103.15.1	S3-HOP15 Fa0/1
	Fa4/0	10.104.24.2	255.255.255.252	10.104.24.0	10.104.24.3	10.104.24.1	S4-HOP24 Fa0/0 (10.104.24.1)

	Fa0/0	202.232.140.10	255.255.255.0	202.232.140.0	202.232.140.255	N/A	Destination Network
	Fa1/0	192.168.20.1	255.255.255.0	192.168.20.0	192.168.20.255	N/A	LAN for client PCs
	Loopback0	10.255.8.1	255.255.255.255	10.255.8.1	10.255.8.1	N/A	Server ID

STUD1 Addressing Table (23 hops)

Name	Interface	IPV4 Address	Subnet Mask	Network Address	Broadcast Address	Gateway	Comment
STUD1							
	S0/0	10.0.1.1	255.255.255.252	10.0.1.0	10.0.1.3	10.0.1.2	Link to FRX1
	Fa1/0	192.168.11.1	255.255.255.0	192.168.11.0	192.168.11.255	N/A	STUD1 LAN Gateway
	Fa0/0	10.101.0.1	255.255.255.252	10.101.0.0	10.101.0.3	10.101.0.2	Link to S1-HOP1
	Tunnel0	172.16.1.1	255.255.255.252	172.16.1.0	172.16.1.3	N/A	GRE VPN to STUD3
	Loopback0	10.255.1.1	255.255.255.255	10.255.1.1	10.255.1.1	N/A	Router ID for eBGP
S1-HOP1							
	Fa0/0	10.101.0.2	255.255.255.252	10.101.0.0	10.101.0.3	10.101.0.1	Link to STUD1
	Fa0/1	10.101.1.1	255.255.255.252	10.101.1.0	10.101.1.3	10.101.1.2	Link to S1-HOP2

S1-HOP2							
	Fa0/1	10.101.1.2	255.255.255.252	10.101.1.0	10.101.1.3	10.101.1.1	Link to S1-HOP1
	Fa0/0	10.101.2.1	255.255.255.252	10.101.2.0	10.101.2.3	10.101.2.2	Link to S1-HOP3
S1-HOP3							
	Fa0/0	10.101.2.2	255.255.255.252	10.101.2.0	10.101.2.3	10.101.2.1	Link to S1-HOP2
	Fa0/1	10.101.3.1	255.255.255.252	10.101.3.0	10.101.3.3	10.101.3.2	Link to S1-HOP4
S1-HOP4							
	Fa0/1	10.101.3.2	255.255.255.252	10.101.3.0	10.101.3.3	10.101.3.1	Link to S1-HOP3
	Fa0/0	10.101.4.1	255.255.255.252	10.101.4.0	10.101.4.3	10.101.4.2	Link to S1-HOP5
S1-HOP5							
	Fa0/0	10.101.4.2	255.255.255.252	10.101.4.0	10.101.4.3	10.101.4.1	Link to S1-HOP4
	Fa0/1	10.101.5.1	255.255.255.252	10.101.5.0	10.101.5.3	10.101.5.2	Link to S1-HOP6
S1-HOP6							
	Fa0/1	10.101.5.2	255.255.255.252	10.101.5.0	10.101.5.3	10.101.5.1	Link to S1-HOP5

	Fa0/0	10.101.6.1	255.255.255.252	10.101.6.0	10.101.6.3	10.101.6.2	Link to S1-HOP7
S1-HOP7							
	Fa0/0	10.101.6.2	255.255.255.252	10.101.6.0	10.101.6.3	10.101.6.1	Link to S1-HOP6
	Fa0/1	10.101.7.1	255.255.255.252	10.101.7.0	10.101.7.3	10.101.7.2	Link to S1-HOP8
S1-HOP8							
	Fa0/1	10.101.7.2	255.255.255.252	10.101.7.0	10.101.7.3	10.101.7.1	Link to S1-HOP7
	Fa0/0	10.101.8.1	255.255.255.252	10.101.8.0	10.101.8.3	10.101.8.2	Link to S1-HOP9
S1-HOP9							
	Fa0/0	10.101.8.2	255.255.255.252	10.101.8.0	10.101.8.3	10.101.8.1	Link to S1-HOP8
	Fa0/1	10.101.9.1	255.255.255.252	10.101.9.0	10.101.9.3	10.101.9.2	Link to S1-HOP10
S1-HOP10							
	Fa0/1	10.101.9.2	255.255.255.252	10.101.9.0	10.101.9.3	10.101.9.1	Link to S1-HOP9
	Fa0/0	10.101.10.1	255.255.255.252	10.101.10.0	10.101.10.3	10.101.10.2	Link to S1-HOP11
S1-HOP11							

	Fa0/0	10.101.10.2	255.255.255.252	10.101.10.0	10.101.10.3	10.101.10.1	Link to S1-HOP10
	Fa0/1	10.101.11.1	255.255.255.252	10.101.11.0	10.101.11.3	10.101.11.2	Link to S1-HOP12
S1-HOP12							
	Fa0/1	10.101.11.2	255.255.255.252	10.101.11.0	10.101.11.3	10.101.11.1	Link to S1-HOP11
	Fa0/0	10.101.12.1	255.255.255.252	10.101.12.0	10.101.12.3	10.101.12.2	Link to S1-HOP13
S1-HOP13							
	Fa0/0	10.101.12.2	255.255.255.252	10.101.12.0	10.101.12.3	10.101.12.1	Link to S1-HOP12
	Fa0/1	10.101.13.1	255.255.255.252	10.101.13.0	10.101.13.3	10.101.13.2	Link to S1-HOP14
S1-HOP14							
	Fa0/1	10.101.13.2	255.255.255.252	10.101.13.0	10.101.13.3	10.101.13.1	Link to S1-HOP13
	Fa0/0	10.101.14.1	255.255.255.252	10.101.14.0	10.101.14.3	10.101.14.2	Link to S1-HOP15
S1-HOP15							
	Fa0/0	10.101.14.2	255.255.255.252	10.101.14.0	10.101.14.3	10.101.14.1	Link to S1-HOP14
	Fa0/1	10.101.15.1	255.255.255.252	10.101.15.0	10.101.15.3	10.101.15.2	Link to S1-HOP16

S1-HOP16							
	Fa0/1	10.101.15.2	255.255.255.252	10.101.15.0	10.101.15.3	10.101.15.1	Link to S1-HOP15
	Fa0/0	10.101.16.1	255.255.255.252	10.101.16.0	10.101.16.3	10.101.16.2	Link to S1-HOP17
S1-HOP17							
	Fa0/0	10.101.16.2	255.255.255.252	10.101.16.0	10.101.16.3	10.101.16.1	Link to S1-HOP16
	Fa0/1	10.101.17.1	255.255.255.252	10.101.17.0	10.101.17.3	10.101.17.2	Link to S1-HOP18
S1-HOP18							
	Fa0/1	10.101.17.2	255.255.255.252	10.101.17.0	10.101.17.3	10.101.17.1	Link to S1-HOP17
	Fa0/0	10.101.18.1	255.255.255.252	10.101.18.0	10.101.18.3	10.101.18.2	Link to S1-HOP19
S1-HOP19							
	Fa0/0	10.101.18.2	255.255.255.252	10.101.18.0	10.101.18.3	10.101.18.1	Link to S1-HOP18
	Fa0/1	10.101.19.1	255.255.255.252	10.101.19.0	10.101.19.3	10.101.19.2	Link to S1-HOP20
S1-HOP20							
	Fa0/1	10.101.19.2	255.255.255.252	10.101.19.0	10.101.19.3	10.101.19.1	Link to S1-HOP19

	Fa0/0	10.101.20.1	255.255.255.252	10.101.20.0	10.101.20.3	10.101.20.2	Link to S1-HOP21
S1-HOP21							
	Fa0/0	10.101.20.2	255.255.255.252	10.101.20.0	10.101.20.3	10.101.20.1	Link to S1-HOP20
	Fa0/1	10.101.21.1	255.255.255.252	10.101.21.0	10.101.21.3	10.101.21.2	Link to S1-HOP22
S1-HOP22							
	Fa0/1	10.101.21.2	255.255.255.252	10.101.21.0	10.101.21.3	10.101.21.1	Link to S1-HOP21
	Fa0/0	10.101.22.1	255.255.255.252	10.101.22.0	10.101.22.3	10.101.22.2	Link to FTPServer (Hop23)

STUD2 Addressing Table (26 hops)

Name	Interface	IPV4 Address	Subnet Mask	Network Address	Broadcast Address	Gateway	Comment
STUD2							
	S0/0	10.0.2.1	255.255.255.252	10.0.2.0	10.0.2.3	10.0.2.2	Link to FRX1
	Fa1/0	192.168.2.1	255.255.255.0	192.168.2.0	192.168.2.255	N/A	STUD2 LAN Gateway
	Fa0/0	10.102.0.1	255.255.255.252	10.102.0.0	10.102.0.3	10.102.0.2	Link to S2-HOP1
	Loopback0	10.255.2.1	255.255.255.255	10.255.2.1	10.255.2.1	N/A	Router ID for eBGP
S2-HOP1							
	Fa0/0	10.102.0.2	255.255.255.252	10.102.0.0	10.102.0.3	10.102.0.1	Link to STUD4
	Fa0/1	10.102.1.1	255.255.255.252	10.102.1.0	10.102.1.3	10.102.1.2	Link to S2-HOP2
S2-HOP2							
	Fa0/1	10.102.1.2	255.255.255.252	10.102.1.0	10.102.1.3	10.102.1.1	Link to S2-HOP1
	Fa0/0	10.102.2.1	255.255.255.252	10.102.2.0	10.102.2.3	10.102.2.2	Link to S2-HOP3
S2-HOP3							

	Fa0/0	10.102.2.2	255.255.255.252	10.102.2.0	10.102.2.3	10.102.2.1	Link to S2-HOP2
	Fa0/1	10.102.3.1	255.255.255.252	10.102.3.0	10.102.3.3	10.102.3.2	Link to S2-HOP4
S2-HOP4							
	Fa0/1	10.102.3.2	255.255.255.252	10.102.3.0	10.102.3.3	10.102.3.1	Link to S2-HOP3
	Fa0/0	10.102.4.1	255.255.255.252	10.102.4.0	10.102.4.3	10.102.4.2	Link to S2-HOP5
S2-HOP5							
	Fa0/0	10.102.4.2	255.255.255.252	10.102.4.0	10.102.4.3	10.102.4.1	Link to S2-HOP4
	Fa0/1	10.102.5.1	255.255.255.252	10.102.5.0	10.102.5.3	10.102.5.2	Link to S2-HOP6
S2-HOP6							
	Fa0/1	10.102.5.2	255.255.255.252	10.102.5.0	10.102.5.3	10.102.5.1	Link to S2-HOP5
	Fa0/0	10.102.6.1	255.255.255.252	10.102.6.0	10.102.6.3	10.102.6.2	Link to S2-HOP7
S2-HOP7							
	Fa0/0	10.102.6.2	255.255.255.252	10.102.6.0	10.102.6.3	10.102.6.1	Link to S2-HOP6
	Fa0/1	10.102.7.1	255.255.255.252	10.102.7.0	10.102.7.3	10.102.7.2	Link to S2-HOP8

S2-HOP8							
	Fa0/1	10.102.7.2	255.255.255.252	10.102.7.0	10.102.7.3	10.102.7.1	Link to S2-HOP7
	Fa0/0	10.102.8.1	255.255.255.252	10.102.8.0	10.102.8.3	10.102.8.2	Link to S2-HOP9
S2-HOP9							
	Fa0/0	10.102.8.2	255.255.255.252	10.102.8.0	10.102.8.3	10.102.8.1	Link to S2-HOP8
	Fa0/1	10.102.9.1	255.255.255.252	10.102.9.0	10.102.9.3	10.102.9.2	Link to S2-HOP10
S2-HOP10							
	Fa0/1	10.102.9.2	255.255.255.252	10.102.9.0	10.102.9.3	10.102.9.1	Link to S2-HOP9
	Fa0/0	10.102.10.1	255.255.255.252	10.102.10.0	10.102.10.3	10.102.10.2	Link to S2-HOP11
S2-HOP11							
	Fa0/0	10.102.10.2	255.255.255.252	10.102.10.0	10.102.10.3	10.102.10.1	Link to S2-HOP10
	Fa0/1	10.102.11.1	255.255.255.252	10.102.11.0	10.102.11.3	10.102.11.2	Link to S2-HOP12
S2-HOP12							
	Fa0/1	10.102.11.2	255.255.255.252	10.102.11.0	10.102.11.3	10.102.11.1	Link to S2-HOP11

	Fa0/0	10.102.12.1	255.255.255.252	10.102.12.0	10.102.12.3	10.102.12.2	Link to S2-HOP13
S2-HOP13							
	Fa0/0	10.102.12.2	255.255.255.252	10.102.12.0	10.102.12.3	10.102.12.1	Link to S2-HOP12
	Fa0/1	10.102.13.1	255.255.255.252	10.102.13.0	10.102.13.3	10.102.13.2	Link to S2-HOP14
S2-HOP14							
	Fa0/1	10.102.13.2	255.255.255.252	10.102.13.0	10.102.13.3	10.102.13.1	Link to S2-HOP13
	Fa0/0	10.102.14.1	255.255.255.252	10.102.14.0	10.102.14.3	10.102.14.2	Link to S2-HOP15
S2-HOP15							
	Fa0/0	10.102.14.2	255.255.255.252	10.102.14.0	10.102.14.3	10.102.14.1	Link to S2-HOP14
	Fa0/1	10.102.15.1	255.255.255.252	10.102.15.0	10.102.15.3	10.102.15.2	Link to S2-HOP16
S2-HOP16***							
	Fa0/1	10.102.15.2	255.255.255.252	10.102.15.0	10.102.15.3	10.102.15.1	Link to S2-HOP15
	Fa0/0	10.102.16.1	255.255.255.252	10.102.16.0	10.102.16.3	10.102.16.2	Link to S2-HOP17
S2-HOP17							

	Fa0/0	10.102.16.2	255.255.255.252	10.102.16.0	10.102.16.3	10.102.16.1	Link to S2-HOP16
	Fa0/1	10.102.17.1	255.255.255.252	10.102.17.0	10.102.17.3	10.102.17.2	Link to S2-HOP18
S2-HOP18							
	Fa0/1	10.102.17.2	255.255.255.252	10.102.17.0	10.102.17.3	10.102.17.1	Link to S2-HOP17
	Fa0/0	10.102.18.1	255.255.255.252	10.102.18.0	10.102.18.3	10.102.18.2	Link to S2-HOP19
S2-HOP19							
	Fa0/0	10.102.18.2	255.255.255.252	10.102.18.0	10.102.18.3	10.102.18.1	Link to S2-HOP18
	Fa0/1	10.102.19.1	255.255.255.252	10.102.19.0	10.102.19.3	10.102.19.2	Link to S2-HOP20
S2-HOP20							
	Fa0/1	10.102.19.2	255.255.255.252	10.102.19.0	10.102.19.3	10.102.19.1	Link to S2-HOP19
	Fa0/0	10.102.20.1	255.255.255.252	10.102.20.0	10.102.20.3	10.102.20.2	Link to S2-HOP21
S2-HOP21							
	Fa0/0	10.102.20.2	255.255.255.252	10.102.20.0	10.102.20.3	10.102.20.1	Link to S2-HOP20
	Fa0/1	10.102.21.1	255.255.255.252	10.102.21.0	10.102.21.3	10.102.21.2	Link to S2-HOP22

S2-HOP22							
	Fa0/1	10.102.21.2	255.255.255.252	10.102.21.0	10.102.21.3	10.102.21.1	Link to S2-HOP21
	Fa0/0	10.102.22.1	255.255.255.252	10.102.22.0	10.102.22.3	10.102.22.2	Link to S2-HOP23
S2-HOP23							
	Fa0/0	10.102.22.2	255.255.255.252	10.102.22.0	10.102.22.3	10.102.22.1	Link to S2-HOP22
	Fa0/1	10.102.23.1	255.255.255.252	10.102.23.0	10.102.23.3	10.102.23.2	Link to S2-HOP24
S2-HOP24							
	Fa0/1	10.102.23.2	255.255.255.252	10.102.23.0	10.102.23.3	10.102.23.1	Link to S2-HOP23
	Fa0/0	10.102.24.1	255.255.255.252	10.102.24.0	10.102.24.3	10.102.24.2	Link to S2-HOP25
S2-HOP25							
	Fa0/1	10.102.24.2	255.255.255.252	10.102.24.0	10.102.24.3	10.102.24.1	Link to S2-HOP24
	Fa0/2	10.102.25.1	255.255.255.252	10.102.25.0	10.102.25.3	10.102.25.2	Link to FTPServer (Hop26)

STUD3 Addressing Table (16 hops)

Name	Interface	IPV4 Address	Subnet Mask	Network Address	Broadcast Address	Gateway	Comment
STUD3							
	S0/3	10.1.1.2	255.255.255.252	10.1.1.0	10.1.1.3	N/A	FR to FR1
	Fa0/0	10.103.0.1*	255.255.255.252	10.103.0.0	10.103.0.3	10.103.0.2	Link to S3-HOP1
	Fa1/0	192.168.3.1	255.255.255.0	192.168.3.0	192.168.3.255	N/A	STUD3 LAN Gateway
	Tunnel0	172.16.1.2*	255.255.255.252	172.16.1.0	172.16.1.3	N/A	GRE VPN to STUD1
	Loopback0	10.255.3.1	255.255.255.255	10.255.3.1	10.255.3.1	N/A	Router ID for eBGP
S3-HOP1							
	Fa0/0	10.103.0.2	255.255.255.252	10.103.0.0	10.103.0.3	10.103.0.1	Link to STUD3
	Fa0/1	10.103.1.1	255.255.255.252	10.103.1.0	10.103.1.3	10.103.1.2	Link to S3-HOP2
S3-HOP2							
	Fa0/1	10.103.1.2	255.255.255.252	10.103.1.0	10.103.1.3	10.103.1.1	Link to S3-HOP1
	Fa0/0	10.103.2.1	255.255.255.252	10.103.2.0	10.103.2.3	10.103.2.2	Link to S3-HOP3

S3-HOP3							
	Fa0/0	10.103.2.2	255.255.255.252	10.103.2.0	10.103.2.3	10.103.2.1	Link to S3-HOP2
	Fa0/1	10.103.3.1	255.255.255.252	10.103.3.0	10.103.3.3	10.103.3.2	Link to S3-HOP4
S3-HOP4							
	Fa0/1	10.103.3.2	255.255.255.252	10.103.3.0	10.103.3.3	10.103.3.1	Link to S3-HOP3
	Fa0/0	10.103.4.1	255.255.255.252	10.103.4.0	10.103.4.3	10.103.4.2	Link to S3-HOP5
S3-HOP5							
	Fa0/0	10.103.4.2	255.255.255.252	10.103.4.0	10.103.4.3	10.103.4.1	Link to S3-HOP4
	Fa0/1	10.103.5.1	255.255.255.252	10.103.5.0	10.103.5.3	10.103.5.2	Link to S3-HOP6
S3-HOP6							
	Fa0/1	10.103.5.2	255.255.255.252	10.103.5.0	10.103.5.3	10.103.5.1	Link to S3-HOP5
	Fa0/0	10.103.6.1	255.255.255.252	10.103.6.0	10.103.6.3	10.103.6.2	Link to S3-HOP7
S3-HOP7							
	Fa0/0	10.103.6.2	255.255.255.252	10.103.6.0	10.103.6.3	10.103.6.1	Link to S3-HOP6

	Fa0/1	10.103.7.1	255.255.255.252	10.103.7.0	10.103.7.3	10.103.7.2	Link to S3-HOP8
S3-HOP8							
	Fa0/1	10.103.7.2	255.255.255.252	10.103.7.0	10.103.7.3	10.103.7.1	Link to S3-HOP7
	Fa0/0	10.103.8.1	255.255.255.252	10.103.8.0	10.103.8.3	10.103.8.2	Link to S3-HOP9
S3-HOP9							
	Fa0/0	10.103.8.2	255.255.255.252	10.103.8.0	10.103.8.3	10.103.8.1	Link to S3-HOP8
	Fa0/1	10.103.9.1	255.255.255.252	10.103.9.0	10.103.9.3	10.103.9.2	Link to S3-HOP10
S3-HOP10							
	Fa0/1	10.103.9.2	255.255.255.252	10.103.9.0	10.103.9.3	10.103.9.1	Link to S3-HOP9
	Fa0/0	10.103.10.1	255.255.255.252	10.103.10.0	10.103.10.3	10.103.10.2	Link to S3-HOP11
S3-HOP11							
	Fa0/0	10.103.10.2	255.255.255.252	10.103.10.0	10.103.10.3	10.103.10.1	Link to S3-HOP10
	Fa0/1	10.103.11.1	255.255.255.252	10.103.11.0	10.103.11.3	10.103.11.2	Link to S3-HOP12
S3-HOP12							

	Fa0/1	10.103.11.2	255.255.255.252	10.103.11.0	10.103.11.3	10.103.11.1	Link to S3-HOP11
	Fa0/0	10.103.12.1	255.255.255.252	10.103.12.0	10.103.12.3	10.103.12.2	Link to S3-HOP13
S3-HOP13							
	Fa0/0	10.103.12.2	255.255.255.252	10.103.12.0	10.103.12.3	10.103.12.1	Link to S3-HOP12
	Fa0/1	10.103.13.1	255.255.255.252	10.103.13.0	10.103.13.3	10.103.13.2	Link to S3-HOP14
S3-HOP14							
	Fa0/1	10.103.13.2	255.255.255.252	10.103.13.0	10.103.13.3	10.103.13.1	Link to S3-HOP13
	Fa0/0	10.103.14.1	255.255.255.252	10.103.14.0	10.103.14.3	10.103.14.2	Link to S3-HOP15
S3-HOP15							
	Fa0/0	10.103.14.2	255.255.255.252	10.103.14.0	10.103.14.3	10.103.14.1	Link to S3-HOP14
	Fa0/3	10.103.15.1	255.255.255.252	10.103.15.0	10.103.15.3	10.103.15.2	Link to FTPServer (Hop16)

STUD4 Addressing Table (25 hops)

Name	Interface	IPV4 Address	Subnet Mask	Network Address	Broadcast Address	Gateway	Comment
STUD4							
	S0/0	10.0.4.1	255.255.255.252	10.0.4.0	10.0.4.3	10.0.4.2	Link to FRX1 (NAT Outside)
	Fa1/0	192.168.4.1	255.255.255.0	192.168.4.0	192.168.4.255	N/A	STUD4 LAN Gateway
	Fa0/0	10.104.0.1	255.255.255.252	10.104.0.0	10.104.0.3	10.104.0.2	Link to S4-HOP1
	Loopback0	10.255.4.1	255.255.255.255	10.255.4.1	10.255.4.1	N/A	Router ID for eBGP
S4-HOP1							
	Fa0/0	10.104.0.2	255.255.255.252	10.104.0.0	10.104.0.3	10.104.0.1	Link to STUD4
	Fa0/1	10.104.1.1	255.255.255.252	10.104.1.0	10.104.1.3	10.104.1.2	Link to S4-HOP2
S4-HOP2							
	Fa0/1	10.104.1.2	255.255.255.252	10.104.1.0	10.104.1.3	10.104.1.1	Link to S4-HOP1
	Fa0/0	10.104.2.1	255.255.255.252	10.104.2.0	10.104.2.3	10.104.2.2	Link to S4-HOP3
S4-HOP3							

	Fa0/0	10.104.2.2	255.255.255.252	10.104.2.0	10.104.2.3	10.104.2.1	Link to S4-HOP2
	Fa0/1	10.104.3.1	255.255.255.252	10.104.3.0	10.104.3.3	10.104.3.2	Link to S4-HOP4
S4-HOP4							
	Fa0/1	10.104.3.2	255.255.255.252	10.104.3.0	10.104.3.3	10.104.3.1	Link to S4-HOP3
	Fa0/0	10.104.4.1	255.255.255.252	10.104.4.0	10.104.4.3	10.104.4.2	Link to S4-HOP5
S4-HOP5							
	Fa0/0	10.104.4.2	255.255.255.252	10.104.4.0	10.104.4.3	10.104.4.1	Link to S4-HOP4
	Fa0/1	10.104.5.1	255.255.255.252	10.104.5.0	10.104.5.3	10.104.5.2	Link to S4-HOP6
S4-HOP6							
	Fa0/1	10.104.5.2	255.255.255.252	10.104.5.0	10.104.5.3	10.104.5.1	Link to S4-HOP5
	Fa0/0	10.104.6.1	255.255.255.252	10.104.6.0	10.104.6.3	10.104.6.2	Link to S4-HOP7
S4-HOP7							
	Fa0/0	10.104.6.2	255.255.255.252	10.104.6.0	10.104.6.3	10.104.6.1	Link to S4-HOP6
	Fa0/1	10.104.7.1	255.255.255.252	10.104.7.0	10.104.7.3	10.104.7.2	Link to S4-HOP8

S4-HOP8							
	Fa0/1	10.104.7.2	255.255.255.252	10.104.7.0	10.104.7.3	10.104.7.1	Link to S4-HOP7
	Fa0/0	10.104.8.1	255.255.255.252	10.104.8.0	10.104.8.3	10.104.8.2	Link to S4-HOP9
S4-HOP9							
	Fa0/0	10.104.8.2	255.255.255.252	10.104.8.0	10.104.8.3	10.104.8.1	Link to S4-HOP8
	Fa0/1	10.104.9.1	255.255.255.252	10.104.9.0	10.104.9.3	10.104.9.2	Link to S4-HOP10
S4-HOP10							
	Fa0/1	10.104.9.2	255.255.255.252	10.104.9.0	10.104.9.3	10.104.9.1	Link to S4-HOP9
	Fa0/0	10.104.10.1	255.255.255.252	10.104.10.0	10.104.10.3	10.104.10.2	Link to S4-HOP11
S4-HOP11							
	Fa0/0	10.104.10.2	255.255.255.252	10.104.10.0	10.104.10.3	10.104.10.1	Link to S4-HOP10
	Fa0/1	10.104.11.1	255.255.255.252	10.104.11.0	10.104.11.3	10.104.11.2	Link to S4-HOP12
S4-HOP12							
	Fa0/1	10.104.11.2	255.255.255.252	10.104.11.0	10.104.11.3	10.104.11.1	Link to S4-HOP11

	Fa0/0	10.104.12.1	255.255.255.252	10.104.12.0	10.104.12.3	10.104.12.2	Link to S4-HOP13
S4-HOP13							
	Fa0/0	10.104.12.2	255.255.255.252	10.104.12.0	10.104.12.3	10.104.12.1	Link to S4-HOP12
	Fa0/1	10.104.13.1	255.255.255.252	10.104.13.0	10.104.13.3	10.104.13.2	Link to S4-HOP14
S4-HOP14							
	Fa0/1	10.104.13.2	255.255.255.252	10.104.13.0	10.104.13.3	10.104.13.1	Link to S4-HOP13
	Fa0/0	10.104.14.1	255.255.255.252	10.104.14.0	10.104.14.3	10.104.14.2	Link to S4-HOP15
S4-HOP15							
	Fa0/0	10.104.14.2	255.255.255.252	10.104.14.0	10.104.14.3	10.104.14.1	Link to S4-HOP14
	Fa0/1	10.104.15.1	255.255.255.252	10.104.15.0	10.104.15.3	10.104.15.2	Link to S4-HOP16
S4-HOP16***							
	Fa0/1	10.104.15.2	255.255.255.252	10.104.15.0	10.104.15.3	10.104.15.1	Link to S4-HOP15
	Fa0/0	10.104.16.1	255.255.255.252	10.104.16.0	10.104.16.3	10.104.16.2	Link to S4-HOP17
S4-HOP17							

	Fa0/0	10.104.16.2	255.255.255.252	10.104.16.0	10.104.16.3	10.104.16.1	Link to S4-HOP16
	Fa0/1	10.104.17.1	255.255.255.252	10.104.17.0	10.104.17.3	10.104.17.2	Link to S4-HOP18
S4-HOP18							
	Fa0/1	10.104.17.2	255.255.255.252	10.104.17.0	10.104.17.3	10.104.17.1	Link to S4-HOP17
	Fa0/0	10.104.18.1	255.255.255.252	10.104.18.0	10.104.18.3	10.104.18.2	Link to S4-HOP19
S4-HOP19							
	Fa0/0	10.104.18.2	255.255.255.252	10.104.18.0	10.104.18.3	10.104.18.1	Link to S4-HOP18
	Fa0/1	10.104.19.1	255.255.255.252	10.104.19.0	10.104.19.3	10.104.19.2	Link to S4-HOP20
S4-HOP20							
	Fa0/1	10.104.19.2	255.255.255.252	10.104.19.0	10.104.19.3	10.104.19.1	Link to S4-HOP19
	Fa0/0	10.104.20.1	255.255.255.252	10.104.20.0	10.104.20.3	10.104.20.2	Link to S4-HOP21
S4-HOP21							
	Fa0/0	10.104.20.2	255.255.255.252	10.104.20.0	10.104.20.3	10.104.20.1	Link to S4-HOP20
	Fa0/1	10.104.21.1	255.255.255.252	10.104.21.0	10.104.21.3	10.104.21.2	Link to S4-HOP22

S4-HOP22							
	Fa0/1	10.104.21.2	255.255.255.252	10.104.21.0	10.104.21.3	10.104.21.1	Link to S4-HOP21
	Fa0/0	10.104.22.1	255.255.255.252	10.104.22.0	10.104.22.3	10.104.22.2	Link to S4-HOP23
S4-HOP23							
	Fa0/0	10.104.22.2	255.255.255.252	10.104.22.0	10.104.22.3	10.104.22.1	Link to S4-HOP22
	Fa0/1	10.104.23.1	255.255.255.252	10.104.23.0	10.104.23.3	10.104.23.2	Link to S4-HOP24
S4-HOP24							
	Fa0/1	10.104.23.2	255.255.255.252	10.104.23.0	10.104.23.3	10.104.23.1	Link to S4-HOP23
	Fa0/4	10.104.24.1	255.255.255.252	10.104.24.0	10.104.24.3	10.104.24.2	Link to FTPServer (Hop25)

Client PCs

Name	Interface	IPV4 Address	Subnet Mask	Network Address	Broadcast Address	Gateway	Comment

SA-PC2 (STUD1-PC)	E0	192.168.11.10	255.255.255.0	192.168.11.0	192.168.11.255	192.168.11.1	DHCP from FTPServer
SA-PC3 (STUD2-PC)	E0	192.168.2.10	255.255.255.0	192.168.2.0	192.168.2.255	192.168.2.1	DHCP from FTPServer
SA-PC41 (STUD3-PC1)	E0	192.168.3.10	255.255.255.0	192.168.3.0	192.168.3.255	192.168.3.1	DHCP from FTPServer
SA-PC42 (STUD3-PC2)	E0	192.168.3.12	255.255.255.0	192.168.3.0	192.168.3.255	192.168.3.1	DHCP from FTPServer
SA-PC51 (STUD4-PC1)	E0	192.168.4.10	255.255.255.0	192.168.4.0	192.168.4.255	192.168.4.1	DHCP from FTPServer
SA-PC52 (STUD4-PC2)	E0	192.168.4.12	255.255.255.0	192.168.4.0	192.168.4.255	192.168.4.1	DHCP from FTPServer
CF-PC1	E0	192.168.5.10	255.255.255.0	192.168.5.0	192.168.5.255	192.168.5.1	DHCP from FTPServer
SA-PC1	E0	192.168.6.10	255.255.255.0	192.168.6.0	192.168.6.255	192.168.6.1	DHCP from FTPServer
FTP-PC1	E0	192.168.20.10	255.255.255.0	192.168.20.0	192.168.20.255	192.168.20.1	

FR Switch	Source Port	Source DLCI	Connected to	Destination Port	Destination DLCI
FR1					
	1	102	STUD3 S0/0	4	201
	2	103	FR2 Port 1	1	301
	3	104	FR3 Port 1	1	401
FR2					
	1	301	FR1 Port 2	2	302
	2	402	FR4 Port 1	1	302
	3	303	FRX1 S0/0	4	501
	4	304	FTP-Server Fa0/0	3	801
FR3					
	1	401	FR1 Port 3	4	104
	2	302	FR4 Port 2	1	402
	3	403	Champs S0/1	4	601

	4	404	STUD2 S0/0	3	901
FR4					
	1	402	FR2 Port 2	2	404
	2	405	St. Aug S0/0	4	701

12 Appendix II – Output (commands)

Configurations

- ***FR1 Configuration***

```
enable
configure terminal
hostname FR1
no ip routing
frame-relay switching
!
map-class frame-relay CIR_STUD3
 frame-relay cir 192000
 frame-relay mincir 192000
 frame-relay bc 1920
 frame-relay be 0
!
map-class frame-relay CIR_128_64
```

```
 frame-relay cir 64000
 frame-relay mincir 64000
 frame-relay bc 640
 frame-relay be 0
!
interface Serial0/3
 description to STUD3
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay intf-type dce
 frame-relay traffic-shaping
 clock rate 256000
 bandwidth 256
 frame-relay class CIR_STUD3
 frame-relay route 100 interface Serial0/1 200
 frame-relay route 101 interface Serial0/2 300
 frame-relay route 102 interface Serial0/1 201
 no shutdown
!
interface Serial0/1
 description to FR2
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay intf-type dce
 frame-relay traffic-shaping
 clock rate 128000
 bandwidth 128
 frame-relay class CIR_128_64
 frame-relay route 200 interface Serial0/3 100
```

```
 frame-relay route 201 interface Serial0/3 102
 no shutdown
!
interface Serial0/2
 description to FR3
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay intf-type dce
 frame-relay traffic-shaping
 clock rate 128000
 bandwidth 128
 frame-relay class CIR_128_64
 frame-relay route 300 interface Serial0/3 101
 no shutdown
!
end
write memory
```

- ***FR2 Configuration***

```
enable
configure terminal
hostname FR2
no ip routing
frame-relay switching
!
```

```
map-class frame-relay CIR_128_64
 frame-relay cir 64000
 frame-relay mincir 64000
 frame-relay bc 640
 frame-relay be 0
!
interface Serial0/1
 description to FR1
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay intf-type dce
 frame-relay traffic-shaping
 clock rate 128000
 bandwidth 128
 frame-relay class CIR_128_64
 frame-relay route 200 interface Serial0/0 500
 frame-relay route 201 interface Serial0/2 400
 no shutdown
!
interface Serial0/0
 description to FRX1
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay intf-type dce
 frame-relay traffic-shaping
 clock rate 128000
 bandwidth 128
 frame-relay class CIR_128_64
 frame-relay route 500 interface Serial0/1 200
```

```
 no shutdown
!
interface Serial0/2
 description to FR4
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay intf-type dce
 frame-relay traffic-shaping
 clock rate 128000
 bandwidth 128
 frame-relay class CIR_128_64
 frame-relay route 400 interface Serial0/1 201
 no shutdown
!
end
write memory
```

- ***FR3 Configuration***

```
enable
configure terminal
hostname FR3
no ip routing
frame-relay switching
!
map-class frame-relay CIR_128_64
 frame-relay cir 64000
 frame-relay mincir 64000
```

```
 frame-relay bc 640
 frame-relay be 0
!
interface Serial0/1
 description to FR1
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay intf-type dce
 frame-relay traffic-shaping
 clock rate 128000
 bandwidth 128
 frame-relay class CIR_128_64
 frame-relay route 300 interface Serial0/0 600
 no shutdown
!
interface Serial0/0
 description to Champs
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay intf-type dce
 frame-relay traffic-shaping
 clock rate 128000
 bandwidth 128
 frame-relay class CIR_128_64
 frame-relay route 600 interface Serial0/1 300
 no shutdown
!
interface Serial0/2
 description to FR4
```

```
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay intf-type dce
 frame-relay traffic-shaping
 clock rate 128000
 bandwidth 128
 frame-relay class CIR_128_64
 no shutdown
!
end
write memory
```

- ***FR4 Configuration***

```
enable
configure terminal
hostname FR4
no ip routing
frame-relay switching
!
map-class frame-relay CIR_128_64
 frame-relay cir 64000
 frame-relay mincir 64000
 frame-relay bc 640
 frame-relay be 0
!
interface Serial0/1
```

```
 description to FR2
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay intf-type dte
 frame-relay traffic-shaping
 bandwidth 128
 frame-relay class CIR_128_64
 frame-relay route 400 interface Serial0/0 800
 no shutdown
!
interface Serial0/0
 description to St. Augustine
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay intf-type dce
 frame-relay traffic-shaping
 clock rate 128000
 bandwidth 128
 frame-relay class CIR_128_64
 frame-relay route 800 interface Serial0/1 400
 no shutdown
!
interface Serial0/2
 description to FR3
 no shutdown
!
end
write memory
```

- ***FRX1 Configuration***

```
enable
configure terminal
hostname FRX1
!
interface Serial0/1
 ip address 10.0.1.2 255.255.255.252
 encapsulation hdlc
 no shutdown
!
interface Serial0/2
 ip address 10.0.2.2 255.255.255.252
 encapsulation hdlc
 no shutdown
!
interface Serial0/3
 ip address 10.0.4.2 255.255.255.252
 encapsulation hdlc
 clock rate 128000
 no shutdown
!
interface Serial0/0
 no ip address
 encapsulation frame-relay
 frame-relay lmi-type ansi
 no shutdown
```

```
!
interface Serial0/0.1 point-to-point
 ip address 10.1.1.1 255.255.255.252
 frame-relay interface-dlci 500
 class CIR_128_64
!
interface FastEthernet0/0
 ip address 202.232.140.1 255.255.255.0
 no shutdown
!
interface Loopback0
 ip address 10.255.7.1 255.255.255.255
!
router bgp 65007
 bgp router-id 10.255.7.1
 neighbor 10.0.1.1 remote-as 65001
 neighbor 10.0.2.1 remote-as 65002
 neighbor 10.0.4.1 remote-as 65004
 neighbor 10.1.1.2 remote-as 65003
 network 10.0.1.0 mask 255.255.255.252
 network 10.0.2.0 mask 255.255.255.252
 network 10.0.4.0 mask 255.255.255.252
 network 10.1.1.0 mask 255.255.255.252
 network 202.232.140.0 mask 255.255.255.0
 network 10.255.7.1 mask 255.255.255.255
!
end
write memory
```

- ***ChampsFleurs Configuration***

```
enable
configure terminal
hostname ChampsFleurs
!
interface Serial0/0
 ip address 10.1.5.2 255.255.255.252
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay map ip 10.1.5.1 600 broadcast
 no shutdown
!
interface FastEthernet1/0
 ip address 192.168.5.1 255.255.255.0
 no shutdown
!
interface Loopback0
 ip address 10.255.5.1 255.255.255.255
!
router bgp 65005
 bgp router-id 10.255.5.1
 neighbor 10.1.10.2 remote-as 65003
 network 10.1.10.0 mask 255.255.255.252
 network 192.168.5.0 mask 255.255.255.0
 network 10.255.5.1 mask 255.255.255.255
```

```
!
access-list 101 permit tcp any host 192.168.4.1 eq 20
access-list 101 permit tcp any host 192.168.4.1 eq 21
access-list 101 permit tcp any host 192.168.4.1 eq 179
access-list 101 permit icmp any host 192.168.4.1
access-list 101 permit udp any host 192.168.4.1 eq 67
access-list 101 permit udp any host 192.168.4.1 eq 68
access-list 101 permit udp any host 192.168.4.1 eq 123
access-list 101 permit tcp any host 192.168.4.1 eq 53
access-list 101 permit udp any host 192.168.4.1 eq 53
access-list 101 permit tcp any host 192.168.4.1 eq 22
access-list 101 deny ip any any
!
interface Serial0/0
 ip access-group 101 in
!
end
write memory
```

- ***StAugustine Configuration***

```
enable
configure terminal
end
hostname StAugustine
!
interface Serial0/0
```

```
 ip address 10.1.6.2 255.255.255.252
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay map ip 10.1.6.1 800 broadcast
 no shutdown
!
interface FastEthernet1/0
 ip address 192.168.6.1 255.255.255.0
 no shutdown
!
interface Loopback0
 ip address 10.255.6.1 255.255.255.255
!
router bgp 65006
 bgp router-id 10.255.6.1
 neighbor 10.1.20.2 remote-as 65003
 network 10.1.20.0 mask 255.255.255.252
 network 192.168.6.0 mask 255.255.255.0
 network 10.255.6.1 mask 255.255.255.255
!
end
write memory
```

- ***STUD1 Configuration***

```
enable
configure terminal
```

```
hostname STUD1
no ip domain-lookup
!
interface Serial0/0
 description Link to FRX1
 ip address 10.0.1.1 255.255.255.252
 encapsulation hdlc
 no shutdown
!
interface FastEthernet0/0
 description Link to S1-HOP1 (STUD1 hop chain)
 ip address 10.101.0.1 255.255.255.252
 no shutdown
!
interface FastEthernet1/0
 description STUD1 LAN Gateway
 ip address 192.168.11.1 255.255.255.0
 no shutdown
!
interface Tunnel0
 description GRE VPN to STUD3
 ip address 172.16.1.1 255.255.255.252
 tunnel source 10.0.1.1
 tunnel destination 10.1.1.2
 keepalive 5 3
!
interface Loopback0
 ip address 10.255.1.1 255.255.255.255
!
```

```
router bgp 65001
 bgp router-id 10.255.1.1
 neighbor 10.0.1.2 remote-as 65007
 neighbor 172.16.1.2 remote-as 65003
 network 10.0.1.0 mask 255.255.255.252
 network 172.16.1.0 mask 255.255.255.252
 network 192.168.11.0 mask 255.255.255.0
 network 10.255.1.1 mask 255.255.255.255
!
! Static routes for hop chain and FTPServer target IP
ip route 10.101.0.0 255.255.0.0 10.101.0.2
ip route 202.232.140.10 255.255.255.255 10.101.22.2
!
end
write memory
```

- ***STUD2 Configuration***

```
enable
configure terminal
hostname STUD2
no ip domain-lookup
!
interface Serial0/0
 description Link to FRX1
 ip address 10.0.2.1 255.255.255.252
 encapsulation hdlc
```

```
 no shutdown
!
interface FastEthernet0/0
 description Link to S2-HOP1 (STUD2 hop chain)
 ip address 10.102.0.1 255.255.255.252
 no shutdown
!
interface FastEthernet1/0
 description STUD2 LAN Gateway
 ip address 192.168.2.1 255.255.255.0
 no shutdown
!
interface Loopback0
 ip address 10.255.2.1 255.255.255.255
!
router bgp 65002
 bgp router-id 10.255.2.1
 neighbor 10.0.2.2 remote-as 65007
 network 10.0.2.0 mask 255.255.255.252
 network 192.168.2.0 mask 255.255.255.0
 network 10.255.2.1 mask 255.255.255.255
!
! Static routes for hop chain and FTPServer target IP
ip route 10.102.0.0 255.255.0.0 10.102.0.2
ip route 202.232.140.10 255.255.255.255 10.102.25.2
!
end
write memory
```

- ***STUD3 Configuration***

```
enable
configure terminal
hostname STUD3
no ip domain-lookup
!
interface Serial0/3
 description FR to FR1 (Frame Relay)
 ip address 10.1.1.2 255.255.255.252
 encapsulation frame-relay
 frame-relay lmi-type ansi
 frame-relay map ip 10.1.1.1 100 broadcast
 frame-relay map ip 10.1.10.1 101 broadcast
 frame-relay map ip 10.1.20.1 102 broadcast
 no shutdown
!
interface FastEthernet0/0
 description Link to S3-HOP1 (STUD3 hop chain)
 ip address 10.103.0.1 255.255.255.252
 no shutdown
!
interface FastEthernet1/0
 description STUD3 LAN Gateway
 ip address 192.168.3.1 255.255.255.0
 no shutdown
!
```

```
interface Tunnel0
 description GRE VPN to STUD1
 ip address 172.16.1.2 255.255.255.252
 tunnel source 10.1.1.2
 tunnel destination 10.0.1.1
 keepalive 5 3
!
interface Loopback0
 ip address 10.255.3.1 255.255.255.255
!
router bgp 65003
 bgp router-id 10.255.3.1
 neighbor 10.1.1.1 remote-as 65007
 neighbor 10.1.10.1 remote-as 65005
 neighbor 10.1.20.1 remote-as 65006
 neighbor 172.16.1.1 remote-as 65001
 network 10.1.1.0 mask 255.255.255.252
 network 10.1.10.0 mask 255.255.255.252
 network 10.1.20.0 mask 255.255.255.252
 network 172.16.1.0 mask 255.255.255.252
 network 192.168.3.0 mask 255.255.255.0
 network 10.255.3.1 mask 255.255.255.255
!
! Static routes for hop chain and FTPServer target IP
ip route 10.103.0.0 255.255.0.0 10.103.0.2
ip route 202.232.140.10 255.255.255.255 10.103.15.2
!
end
write memory
```

- ***STUD4 Configuration***

```
enable
configure terminal
hostname STUD4
no ip domain-lookup
!
interface Serial0/0
 description Link to FRX1
 ip address 10.0.4.1 255.255.255.252
 encapsulation hdlc
 ip nat outside
 no shutdown
!
interface FastEthernet0/0
 description Link to S4-HOP1
 ip address 10.104.0.1 255.255.255.252
 no shutdown
!
interface FastEthernet1/0
 description STUD4 LAN Gateway
 ip address 192.168.4.1 255.255.255.0
 ip nat inside
 no shutdown
!
```

```
interface Loopback0
 ip address 10.255.4.1 255.255.255.255
!
access-list 1 permit 192.168.4.0 0.0.0.255
ip nat inside source list 1 interface Serial0/0 overload
!
router bgp 65004
 bgp router-id 10.255.4.1
 neighbor 10.0.4.2 remote-as 65007
 network 10.0.4.0 mask 255.255.255.252
 network 192.168.4.0 mask 255.255.255.0
 network 10.255.4.1 mask 255.255.255.255
!
ip route 10.104.0.0 255.255.0.0 10.104.0.2
ip route 202.232.140.10 255.255.255.255 10.104.24.2
!
end
write memory
```

- ***FTPServer Configuration***

```
enable
configure terminal
hostname FTPServer
!
```

```
interface FastEthernet0/0
 ip address 202.232.140.10 255.255.255.0
 no shutdown
!
interface FastEthernet3/0
 ip address 10.103.15.2 255.255.255.252
 no shutdown
!
interface FastEthernet1/0
 ip address 192.168.20.1 255.255.255.0
 no shutdown
!
interface Loopback0
 ip address 10.255.8.1 255.255.255.255
!
ip route 0.0.0.0 0.0.0.0 10.103.15.1
!
ip route 10.101.0.0 255.255.0.0 10.101.22.1 ! STUD1 hops
ip route 10.102.0.0 255.255.0.0 10.102.25.1 ! STUD2 hops
ip route 10.103.0.0 255.255.0.0 10.103.15.1 ! STUD3 hops
ip route 10.104.0.0 255.255.0.0 10.104.24.1 ! STUD4 hops

ip route 192.168.11.0 255.255.255.0 10.101.22.1  ! STUD1 LAN
ip route 192.168.2.0 255.255.255.0 10.102.25.1    ! STUD2 LAN
ip route 192.168.3.0 255.255.255.0 10.103.15.1    ! STUD3 LAN
ip route 192.168.4.0 255.255.255.0 10.104.24.1    ! STUD4 LAN

!
! DHCP excluded addresses
```

```
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp excluded-address 192.168.2.1 192.168.2.10
ip dhcp excluded-address 192.168.3.1 192.168.3.10
ip dhcp excluded-address 192.168.4.1 192.168.4.10
ip dhcp excluded-address 192.168.5.1 192.168.5.10
ip dhcp excluded-address 192.168.6.1 192.168.6.10
ip dhcp excluded-address 192.168.11.1 192.168.11.10
ip dhcp pool STUD1_LAN
 network 192.168.11.0 255.255.255.0
 default-router 192.168.11.1
 dns-server 202.232.140.10
!
ip dhcp pool STUD2_LAN
 network 192.168.2.0 255.255.255.0
 default-router 192.168.2.1
 dns-server 202.232.140.10
!
ip dhcp pool STUD3_LAN
 network 192.168.3.0 255.255.255.0
 default-router 192.168.3.1
 dns-server 202.232.140.10
!
ip dhcp pool STUD4_LAN
 network 192.168.4.0 255.255.255.0
 default-router 192.168.4.1
 dns-server 202.232.140.10
!
ip dhcp pool Champs_LAN
 network 192.168.5.0 255.255.255.0
```

```
 default-router 192.168.5.1
 dns-server 202.232.140.10
!
ip dhcp pool StAug_LAN
 network 192.168.6.0 255.255.255.0
 default-router 192.168.6.1
 dns-server 202.232.140.10
!
ntp master 1
ip dns server
ip name-server 8.8.8.8
!
! (optional static host mappings)
ip host STUD1 192.168.11.1
ip host STUD2 192.168.2.1
ip host STUD3 192.168.3.1
ip host STUD4 192.168.4.1
ip host ChampsFleurs 192.168.5.1
ip host StAugustine 192.168.6.1
ip host FRX1 10.255.7.1
ip host FTPServer 202.232.140.10
!
end
write memory
```

STUD1 HOP CHAIN CONFIGURATIONS

- **S1-HOP1 Configuration**

```
enable
configure terminal
hostname S1-HOP1
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.101.0.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.101.1.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP2 Configuration**

```
enable
configure terminal
hostname S1-HOP2
no ip domain-lookup
!
```

```
interface FastEthernet0/1
 ip address 10.101.1.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.101.2.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP3 Configuration**

```
enable
configure terminal
hostname S1-HOP3
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.101.2.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.101.3.1 255.255.255.252
 no shutdown
```

```
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP4 Configuration**

```
enable
configure terminal
hostname S1-HOP4
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.101.3.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.101.4.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP5 Configuration**

```
enable
configure terminal
hostname S1-HOP5
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.101.4.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.101.5.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP6 Configuration**

```
enable
configure terminal
hostname S1-HOP6
```

```
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.101.5.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.101.6.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP7 Configuration**

```
enable
configure terminal
hostname S1-HOP7
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.101.6.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
```

```
 ip address 10.101.7.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP8 Configuration**

```
enable
configure terminal
hostname S1-HOP8
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.101.7.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.101.8.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
```

```
!
end
write memory
```

- **S1-HOP9 Configuration**

```
enable
configure terminal
hostname S1-HOP9
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.101.8.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.101.9.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP10 Configuration**

```
enable
configure terminal
hostname S1-HOP10
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.101.9.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.101.10.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP11 Configuration**

```
enable
configure terminal
hostname S1-HOP11
no ip domain-lookup
!
```

```
interface FastEthernet0/0
 ip address 10.101.10.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.101.11.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP12 Configuration**

```
enable
configure terminal
hostname S1-HOP12
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.101.11.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.101.12.1 255.255.255.252
```

```
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP13 Configuration**

```
enable
configure terminal
hostname S1-HOP13
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.101.12.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.101.13.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
```

```
end
write memory
```

- **S1-HOP14 Configuration**

```
enable
configure terminal
hostname S1-HOP14
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.101.13.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.101.14.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP15 Configuration**

```
enable
configure terminal
hostname S1-HOP15
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.101.14.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.101.15.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP16 Configuration**

```
enable
configure terminal
hostname S1-HOP16
no ip domain-lookup
!
interface FastEthernet0/1
```

```
 ip address 10.101.15.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.101.16.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP17 Configuration**

```
enable
configure terminal
hostname S1-HOP17
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.101.16.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.101.17.1 255.255.255.252
 no shutdown
```

```
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP18 Configuration**

```
enable
configure terminal
hostname S1-HOP18
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.101.17.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.101.18.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
```

```
write memory
```

- **S1-HOP19 Configuration**

```
enable
configure terminal
hostname S1-HOP19
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.101.18.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.101.19.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP20 Configuration**

```
enable
```

```
configure terminal
hostname S1-HOP20
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.101.19.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.101.20.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP21 Configuration**

```
enable
configure terminal
hostname S1-HOP21
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.101.20.2 255.255.255.252
```

```
 no shutdown
!
interface FastEthernet0/1
 ip address 10.101.21.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S1-HOP22 (last hop) Configuration**

```
enable
configure terminal
hostname S1-HOP22
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.101.21.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.101.22.1 255.255.255.252
 no shutdown
!
```

```
router eigrp 100
 network 10.0.0.0
 no auto-summary
exit
router eigrp 100
 redistribute static
 default-metric 10000 100 255 1 1500
exit
!
ip route 0.0.0.0 0.0.0.0 10.101.22.2
!
end
write memory
```

STUD2 HOP CHAIN CONFIGURATIONS

- **S2-HOP1 Configuration**

```
enable
configure terminal
hostname S2-HOP1
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.0.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.102.1.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP2 Configuration**

```
enable
configure terminal
```

```
hostname S2-HOP2
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.102.1.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.102.2.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP3 Configuration**

```
enable
configure terminal
hostname S2-HOP3
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.2.2 255.255.255.252
 no shutdown
```

```
!
interface FastEthernet0/1
 ip address 10.102.3.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP4 Configuration**

```
enable
configure terminal
hostname S2-HOP4
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.102.3.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.102.4.1 255.255.255.252
 no shutdown
!
router eigrp 100
```

```
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP5 Configuration**

```
enable
configure terminal
hostname S2-HOP5
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.4.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.102.5.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP6 Configuration**

```
enable
configure terminal
hostname S2-HOP6
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.102.5.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.102.6.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP7 Configuration**

```
enable
configure terminal
hostname S2-HOP7
```

```
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.6.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.102.7.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP8 Configuration**

```
enable
configure terminal
hostname S2-HOP8
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.102.7.2 255.255.255.252
 no shutdown
!
```

```
interface FastEthernet0/0
 ip address 10.102.8.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP9 Configuration**

```
enable
configure terminal
hostname S2-HOP9
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.8.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.102.9.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
```

```
 no auto-summary
!
end
write memory
```

- **S2-HOP10 Configuration**

```
enable
configure terminal
hostname S2-HOP10
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.102.9.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.102.10.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP11 Configuration**

```
enable
configure terminal
hostname S2-HOP11
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.10.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.102.11.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP12 Configuration**

```
enable
configure terminal
hostname S2-HOP12
no ip domain-lookup
```

```
!
interface FastEthernet0/1
 ip address 10.102.11.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.102.12.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP13 Configuration**

```
enable
configure terminal
hostname S2-HOP13
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.12.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
```

```
 ip address 10.102.13.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP14 Configuration**

```
enable
configure terminal
hostname S2-HOP14
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.102.13.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.102.14.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
```

```
!
end
write memory
```

- **S2-HOP15 Configuration**

```
enable
configure terminal
hostname S2-HOP15
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.14.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.102.15.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP16 Configuration**

```
enable
configure terminal
hostname S2-HOP16
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.102.15.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.102.16.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP17 Configuration**

```
enable
configure terminal
hostname S2-HOP17
no ip domain-lookup
!
```

```
interface FastEthernet0/0
 ip address 10.102.16.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.102.17.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP18 Configuration**

```
enable
configure terminal
hostname S2-HOP18
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.102.17.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.102.18.1 255.255.255.252
```

```
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP19 Configuration**

```
enable
configure terminal
hostname S2-HOP19
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.18.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.102.19.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
```

```
end
write memory
```

- **S2-HOP20 Configuration**

```
enable
configure terminal
hostname S2-HOP20
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.102.19.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.102.20.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP21 Configuration**

```
enable
configure terminal
hostname S2-HOP21
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.20.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.102.21.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP22 Configuration**

```
enable
configure terminal
hostname S2-HOP22
no ip domain-lookup
!
interface FastEthernet0/1
```

```
 ip address 10.102.21.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.102.22.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP23 Configuration**

```
enable
configure terminal
hostname S2-HOP23
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.22.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.102.23.1 255.255.255.252
 no shutdown
```

```
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S2-HOP24 Configuration**

```
enable
configure terminal
hostname S2-HOP24
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.102.23.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.102.24.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
```

```
write memory
```

- **S2-HOP25 (last hop) Configuration**

```
enable
configure terminal
hostname S2-HOP25
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.102.24.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.102.25.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 202.232.140.10 255.255.255.255 10.102.25.2
ip route 0.0.0.0 0.0.0.0 10.102.25.2
router eigrp 100
 redistribute static
 default-metric 10000 100 255 1 1500
!
end
```

```
write memory
```

STUD3 HOP CHAIN CONFIGURATIONS

- **S3-HOP1 Configuration**

```
enable
configure terminal
hostname S3-HOP1
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.103.0.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.103.1.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP2 Configuration**

```
enable
configure terminal
```

```
hostname S3-HOP2
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.103.1.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.103.2.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP3 Configuration**

```
enable
configure terminal
hostname S3-HOP3
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.103.2.2 255.255.255.252
 no shutdown
```

```
!
interface FastEthernet0/1
 ip address 10.103.3.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP4 Configuration**

```
enable
configure terminal
hostname S3-HOP4
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.103.3.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.103.4.1 255.255.255.252
 no shutdown
!
router eigrp 100
```

```
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP5 Configuration**

```
enable
configure terminal
hostname S3-HOP5
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.103.4.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.103.5.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP6 Configuration**

```
enable
configure terminal
hostname S3-HOP6
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.103.5.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.103.6.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP7 Configuration**

```
enable
configure terminal
hostname S3-HOP7
```

```
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.103.6.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.103.7.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP8 Configuration**

```
enable
configure terminal
hostname S3-HOP8
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.103.7.2 255.255.255.252
 no shutdown
!
```

```
interface FastEthernet0/0
 ip address 10.103.8.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP9 Configuration**

```
enable
configure terminal
hostname S3-HOP9
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.103.8.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.103.9.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
```

```
 no auto-summary
!
end
write memory
```

- **S3-HOP10 Configuration**

```
enable
configure terminal
hostname S3-HOP10
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.103.9.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.103.10.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP11 Configuration**

```
enable
configure terminal
hostname S3-HOP11
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.103.10.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.103.11.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP12 Configuration**

```
enable
configure terminal
hostname S3-HOP12
no ip domain-lookup
```

```
!
interface FastEthernet0/1
 ip address 10.103.11.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.103.12.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP13 Configuration**

```
enable
configure terminal
hostname S3-HOP13
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.103.12.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
```

```
 ip address 10.103.13.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
end
write memory
```

- **S3-HOP14 Configuration**

```
enable
configure terminal
hostname S3-HOP14
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.103.13.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.103.14.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
```

```
!
end
```

- **S3-HOP15 Configuration**

```
enable
configure terminal
hostname S3-HOP15
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.103.14.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.103.15.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 0.0.0.0 0.0.0.0 10.103.15.2
!
end
write memory
```

STUD4 HOP CHAIN CONFIGURATIONS

- **S4-HOP1 Configuration**

```
enable
configure terminal
hostname S4-HOP1
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.104.0.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.104.1.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.0.1
ip route 202.232.140.10 255.255.255.255 10.104.1.2
!
end
write memory
```

- **S4-HOP2 Configuration**

```
enable
configure terminal
hostname S4-HOP2
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.104.1.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.2.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.1.1
ip route 202.232.140.10 255.255.255.255 10.104.2.2
!
end
write memory
```

- **S4-HOP3 Configuration**

```
enable
configure terminal
```

```
hostname S4-HOP3
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.104.2.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.104.3.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.2.1
ip route 202.232.140.10 255.255.255.255 10.104.3.2
!
end
write memory
```

- **S4-HOP4 Configuration**

```
enable
configure terminal
hostname S4-HOP4
no ip domain-lookup
!
```

```
interface FastEthernet0/1
 ip address 10.104.3.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.4.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.3.1
ip route 202.232.140.10 255.255.255.255 10.104.4.2
!
end
write memory
```

- **S4-HOP5 Configuration**

```
enable
configure terminal
hostname S4-HOP5
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.104.4.2 255.255.255.252
 no shutdown
```

```
!
interface FastEthernet0/1
 ip address 10.104.5.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.4.1
ip route 202.232.140.10 255.255.255.255 10.104.5.2
!
end
write memory
```

- **S4-HOP6 Configuration**

```
enable
configure terminal
hostname S4-HOP6
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.104.5.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.6.1 255.255.255.252
```

```
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.5.1
ip route 202.232.140.10 255.255.255.255 10.104.6.2
!
end
write memory
```

- **S4-HOP7 Configuration**

```
enable
configure terminal
hostname S4-HOP7
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.104.6.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.104.7.1 255.255.255.252
 no shutdown
!
router eigrp 100
```

```
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.6.1
ip route 202.232.140.10 255.255.255.255 10.104.7.2
!
end
write memory
```

- **S4-HOP8 Configuration**

```
enable
configure terminal
hostname S4-HOP8
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.104.7.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.8.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
```

```
ip route 192.168.4.0 255.255.255.0 10.104.7.1
ip route 202.232.140.10 255.255.255.255 10.104.8.2
!
end
write memory
```

- **S4-HOP9 Configuration**

```
enable
configure terminal
hostname S4-HOP9
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.104.8.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.104.9.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.8.1
ip route 202.232.140.10 255.255.255.255 10.104.9.2
!
```

```
end
write memory
```

- **S4-HOP10 Configuration**

```
enable
configure terminal
hostname S4-HOP10
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.104.9.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.10.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.9.1
ip route 202.232.140.10 255.255.255.255 10.104.10.2
!
end
write memory
```

- **S4-HOP11 Configuration**

```
enable
configure terminal
hostname S4-HOP11
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.104.10.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.104.11.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.10.1
ip route 202.232.140.10 255.255.255.255 10.104.11.2
!
end
write memory
```

- **S4-HOP12 Configuration**

```
enable
configure terminal
hostname S4-HOP12
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.104.11.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.12.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.11.1
ip route 202.232.140.10 255.255.255.255 10.104.12.2
!
end
write memory
```

- **S4-HOP13 Configuration**

```
enable
configure terminal
hostname S4-HOP13
```

```
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.104.12.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.104.13.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.12.1
ip route 202.232.140.10 255.255.255.255 10.104.13.2
!
end
write memory
```

- **S4-HOP14 Configuration**

```
enable
configure terminal
hostname S4-HOP14
no ip domain-lookup
!
interface FastEthernet0/1
```

```
 ip address 10.104.13.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.14.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.13.1
ip route 202.232.140.10 255.255.255.255 10.104.14.2
!
end
write memory
```

- **S4-HOP15 Configuration**

```
enable
configure terminal
hostname S4-HOP15
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.104.14.2 255.255.255.252
 no shutdown
!
```

```
interface FastEthernet0/1
 ip address 10.104.15.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.14.1
ip route 202.232.140.10 255.255.255.255 10.104.15.2
!
end
write memory
```

- **S4-HOP16 Configuration**

```
enable
configure terminal
hostname S4-HOP16
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.104.15.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.16.1 255.255.255.252
 no shutdown
```

```
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.15.1
ip route 202.232.140.10 255.255.255.255 10.104.16.2
!
end
write memory
```

- **S4-HOP17 Configuration**

```
enable
configure terminal
hostname S4-HOP17
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.104.16.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.104.17.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
```

```
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.16.1
ip route 202.232.140.10 255.255.255.255 10.104.17.2
!
end
write memory
```

- **S4-HOP18 Configuration**

```
enable
configure terminal
hostname S4-HOP18
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.104.17.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.18.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.17.1
```

```
ip route 202.232.140.10 255.255.255.255 10.104.18.2
!
end
write memory
```

- **S4-HOP19 Configuration**

```
enable
configure terminal
hostname S4-HOP19
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.104.18.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.104.19.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.18.1
ip route 202.232.140.10 255.255.255.255 10.104.19.2
!
end
```

```
write memory
```

- **S4-HOP20 Configuration**

```
enable
configure terminal
hostname S4-HOP20
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.104.19.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.20.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.19.1
ip route 202.232.140.10 255.255.255.255 10.104.20.2
!
end
write memory
```

- **S4-HOP21 Configuration**

```
enable
configure terminal
hostname S4-HOP21
no ip domain-lookup
!
interface FastEthernet0/0
 ip address 10.104.20.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.104.21.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.20.1
ip route 202.232.140.10 255.255.255.255 10.104.21.2
!
end
write memory
```

- **S4-HOP22 Configuration**

```
enable
```

```
configure terminal
hostname S4-HOP22
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.104.21.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.22.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.21.1
ip route 202.232.140.10 255.255.255.255 10.104.22.2
!
end
write memory
```

- **S4-HOP23 Configuration**

```
enable
configure terminal
hostname S4-HOP23
no ip domain-lookup
```

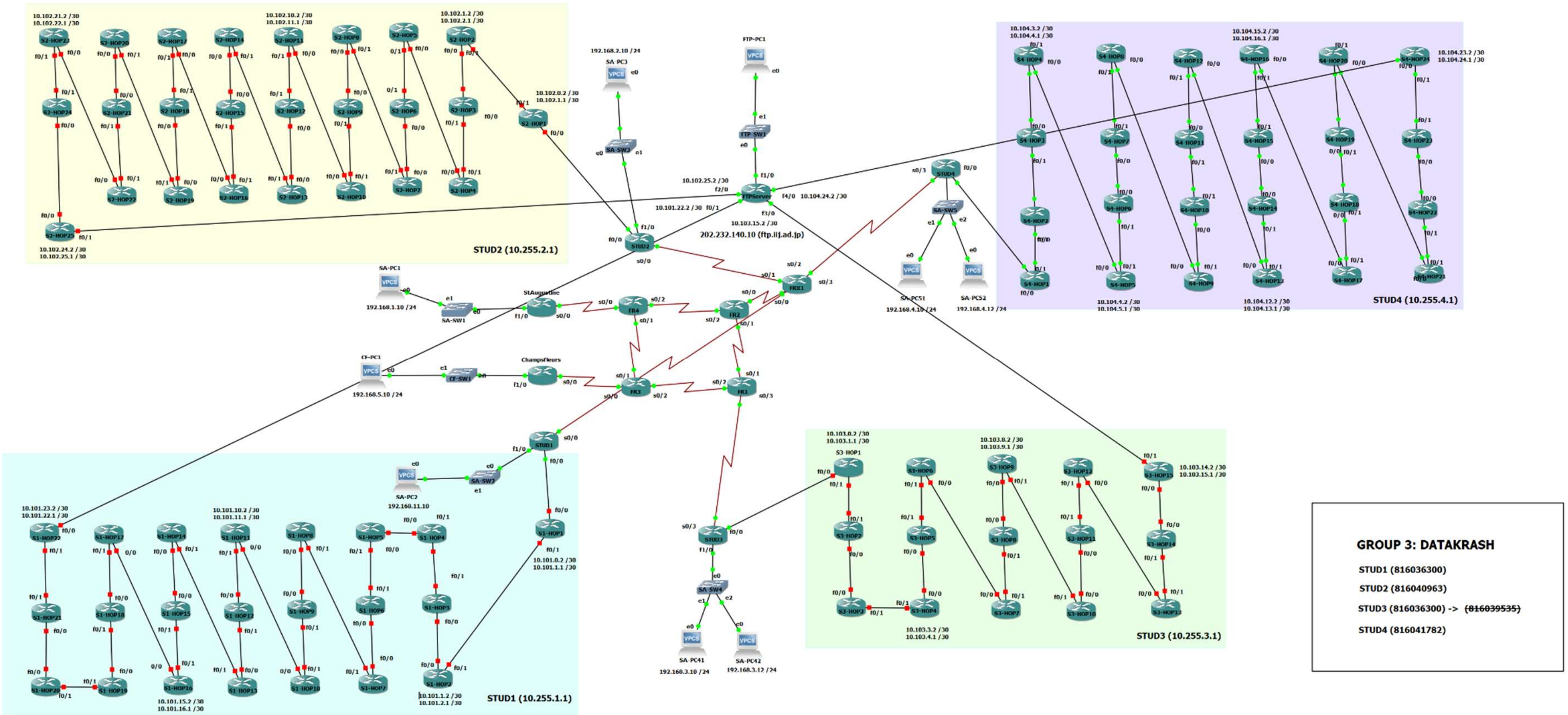
```
!
interface FastEthernet0/0
 ip address 10.104.22.2 255.255.255.252
 no shutdown
!
interface FastEthernet0/1
 ip address 10.104.23.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.22.1
ip route 202.232.140.10 255.255.255.255 10.104.23.2
!
end
write memory
```

- **S4-HOP24 Configuration**

```
enable
configure terminal
hostname S4-HOP24
no ip domain-lookup
!
interface FastEthernet0/1
 ip address 10.104.23.2 255.255.255.252
```

```
 no shutdown
!
interface FastEthernet0/0
 ip address 10.104.24.1 255.255.255.252
 no shutdown
!
router eigrp 100
 network 10.0.0.0
 no auto-summary
!
ip route 192.168.4.0 255.255.255.0 10.104.23.1
ip route 202.232.140.10 255.255.255.255 10.104.24.2
!
end
write memory
```

GNS3 Topology



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